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Compound for organic electric element, organic electric element using same, and electronic apparatus thereof

Abstract

Provided are an organic electronic element comprising a light emitting layer composed of a mixture of compounds capable of improving luminous efficiency, stability, and lifespan of the element, and an electronic device therefor.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2013/0341602	12/2012	Hikime et al.	N/A	N/A
2019/0312215	12/2018	Kang	N/A	H10K 85/615

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2012049518	12/2011	JP	N/A
2013-131518	12/2012	JP	N/A
2013-243266	12/2012	JP	N/A
10-1170666	12/2011	KR	N/A
10-2016-0030402	12/2015	KR	N/A
10-2018-0031874	12/2017	KR	N/A
2016/015815	12/2015	WO	N/A

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Background/Summary

BACKGROUND

Technical Field

(1) The present invention relates to compound for organic electric element, organic electric element using the same, and an electronic device thereof.

Background Art

(2) In general, organic light emitting phenomenon refers to a phenomenon that converts electric energy into light energy by using an organic material. An organic electric element using an organic

light emitting phenomenon usually has a structure including an anode, a cathode, and an organic material layer interposed therebetween. Here, in order to increase the efficiency and stability of the organic electric element, the organic material layer is often composed of a multi-layered structure composed of different materials, and for example, may include a hole injection layer, a hole transport layer, an emitting layer, an electron transport layer, an electron injection layer and the like.

(3) A material used as an organic material layer in an organic electric element may be classified into a light emitting material and a charge transport material, such as a hole injection material, a hole transport material, an electron transport material, an electron injection material and the like depending on its function.

(4) Bis-type cyclic compounds including heteroatoms have a very large difference in properties depending on the material structure, and are therefore applied to various layers as materials for organic electric elements. In particular, the band gap (HOMO, LUMO), electrical properties, chemical properties, and physical properties are different depending on the number of rings and the fused position, and the type and arrangement of heteroatoms, therefore application development for layers of various organic electric elements using the same has been progressed. In a phosphorescent organic electric element using a phosphorescent dopant material, the LUMO and HOMO levels of the host material have a great influence on the efficiency and life span of the organic electric element, and depending on whether electron and hole injection in the emission layer can be efficiently controlled, charge balance in the emission layer, dopant quenching, and reduction in efficiency and lifespan due to light emission at the hole transport layer interface can be prevented.

(5) For fluorescent and phosphorescent host materials, recently we have been studying the increase of efficiency and life span of organic electric elements using TADF (thermal activated delayed fluorescent), exciplex, etc., particularly, and many studies have been carried out to identify the energy transfer method from the host material to the dopant material.

(6) Although there are various methods for identifying the energy transfer in the emitting layer for TADF (thermally activated delayed fluorescent) and exciplex, it can be easily confirmed by the PL lifetime (TRTP) measurement method.

(7) The TRTP (Time Resolved Transient PL) measurement method is a method of observing a decay time after irradiating a pulsed light source onto a host thin film, and is a measurement method that can identify the energy transfer method by observing energy transfer and emission delay time. The TRTP measurement is a measurement method capable of distinguishing fluorescence and phosphorescence, and an energy transfer method in a mixed host material, an exciplex energy transfer method, and a TADF energy transfer method.

(8) As such, there are various factors that affect the efficiency and lifespan depending on how energy is transferred from the host material to the dopant material.

(9) Since the energy transfer method is different depending on the material, the development of a stable and efficient host material for an organic electric element has not been sufficiently performed. Therefore, development of new materials is continuously required, and especially development of a host material for an emitting layer is urgently required.

(10) Reference KR101170666 B1 was used as a prior art document.

DETAILED DESCRIPTION OF THE INVENTION

Summary

(11) The present invention has been proposed in order to solve the problems of the phosphorescent host material, and an object of the present invention is, by controlling the HOMO level of a host material of a phosphorescent emitting organic electric element including a phosphorescent dopant, to provide a compound capable of controlling charge balance and of improving efficiency and lifespan in an emitting layer, and an organic electric element using the same and an electronic device thereof.

Technical Solution

(12) In order to control the efficient hole injection in the emitting layer of the phosphorescent emitting organic electric element, by containing a second host material in combination with a first host material as a main component, the energy barrier between the emitting layer and the adjacent layer can be reduced, and the charge balance in the emitting layer is maximized to provide high efficiency and high lifespan of the organic electric element.

(13) The present invention provides an organic electronic element comprising a first electrode, a second electrode, and an organic material layer formed between the first electrode and the second electrode, wherein the organic material layer comprises an emitting layer, wherein the emitting layer is a phosphorescent emitting layer and comprises a first host compound represented by Formula 1 and a second host compound represented by Formula 2.

(14) ##STR00001##

(15) The present invention also provides organic electric elements and electronic devices using the compounds represented by the Formulas.

Effects of the Invention

(16) By using the mixture according to the present invention as a phosphorescent host material, it is possible to achieve a high luminous efficiency and a low driving voltage of an organic electric element, and the life span of the device can be greatly improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The FIGURE is an illustration of an organic electroluminescent device according to the present invention.

(2) TABLE-US-00001 100: organic electric element, 110: substrate 120: the first electrode(anode), 130: the hole injection layer 140: the hole transport layer, 141: a buffer layer 150: the emitting layer, 151: the emitting auxiliary layer 160: the electron transport layer, 170: the electron injection layer 180: the second electrode(cathode)

DETAILED DESCRIPTION

(3) Hereinafter, some embodiments of the present invention will be described in detail. Further, in the following description of the present invention, a detailed description of known functions and configurations incorporated herein will be omitted when it may make the subject matter of the present invention rather unclear.

(4) In addition, terms, such as first, second, A, B, (a), (b) or the like may be used herein when describing components of the present invention. Each of these terminologies is not used to define an essence, order or sequence of a corresponding component but used merely to distinguish the corresponding component from other component(s). It should be noted that if a component is described as being “connected”, “coupled”, or “connected” to another component, the component may be directly connected or connected to the other component, but another component may be “connected”, “coupled” or “connected” between each component.

(5) As used in the specification and the accompanying claims, unless otherwise stated, the following is the meaning of the term as follows.

(6) Unless otherwise stated, the term “halo” or “halogen”, as used herein, includes fluorine, bromine, chlorine, or iodine.

(7) Unless otherwise stated, the term “alkyl” or “alkyl group”, as used herein, has a single bond of 1 to 60 carbon atoms, and means saturated aliphatic functional radicals including a linear alkyl group, a branched chain alkyl group, a cycloalkyl group (alicyclic), an cycloalkyl group substituted with a alkyl or an alkyl group substituted with a cycloalkyl.

(8) Unless otherwise stated, the term “haloalkyl” or “halogen alkyl”, as used herein, includes an alkyl group substituted with a halogen.

- (9) Unless otherwise stated, the term “heteroalkyl”, as used herein, means alkyl substituted one or more of carbon atoms consisting of an alkyl with hetero atom.
- (10) Unless otherwise stated, the term “alkenyl” or “alkynyl”, as used herein, has double or triple bonds of 2 to 60 carbon atoms, but is not limited thereto, and includes a linear or a branched chain group.
- (11) Unless otherwise stated, the term “cycloalkyl”, as used herein, means alkyl forming a ring having 3 to 60 carbon atoms, but is not limited thereto.
- (12) Unless otherwise stated, the term “alkoxyl group”, “alkoxy group” or “alkyloxy group”, as used herein, means an oxygen radical attached to an alkyl group, but is not limited thereto, and has 1 to 60 carbon atoms.
- (13) Unless otherwise stated, the term “alkenoxyl group”, “alkenoxy group”, “alkenyloxyl group” or “alkenyloxy group”, as used herein, means an oxygen radical attached to an alkenyl group, but is not limited thereto, and has 2 to 60 carbon atoms.
- (14) Unless otherwise stated, the term “aryloxyl group” or “aryloxy group”, as used herein, means an oxygen radical attached to an aryl group, but is not limited thereto, and has 6 to 60 carbon atoms.
- (15) Unless otherwise stated, the term “aryl group” or “arylene group”, as used herein, has 6 to 60 carbon atoms, but is not limited thereto. Herein, the aryl group or arylene group means a monocyclic and polycyclic aromatic group, and may also be formed in conjunction with an adjacent group. Examples of “aryl group” may include a phenyl group, a biphenyl group, a fluorene group, or a spirofluorene group.
- (16) The prefix “aryl” or “ar” means a radical substituted with an aryl group. For example, an arylalkyl may be an alkyl substituted with an aryl, and an arylalenyl may be an alkenyl substituted with aryl, and a radical substituted with an aryl has a number of carbon atoms as defined herein.
- (17) Also, when prefixes are named subsequently, it means that substituents are listed in the order described first. For example, an arylalkoxy means an alkoxy substituted with an aryl, an alkoxylcarbonyl means a carbonyl substituted with an alkoxyl, and an arylcarbonylalkenyl also means an alkenyl substituted with an arylcarbonyl, wherein the arylcarbonyl may be a carbonyl substituted with an aryl.
- (18) Unless otherwise stated, the term “heteroalkyl”, as used herein, means alkyl containing one or more of hetero atoms. Unless otherwise stated, the term “heteroaryl group” or “heteroarylene group”, as used herein, means a C2 to C60 aryl containing one or more of hetero atoms or arylene group, but is not limited thereto, and includes at least one of monocyclic and polycyclic rings, and may also be formed in conjunction with an adjacent group.
- (19) Unless otherwise stated, the term “heterocyclic group”, as used herein, contains one or more heteroatoms, but is not limited thereto, has 2 to 60 carbon atoms, includes any one of monocyclic and polycyclic rings, and may include heteroaliphatic ring and/or heteroaromatic ring. Also, the heterocyclic group may also be formed in conjunction with an adjacent group.
- (20) Unless otherwise stated, the term “heteroatom”, as used herein, represents at least one of N, O, S, P, or Si.
- (21) Also, the term “heterocyclic group” may include a ring containing SO₂ instead of carbon consisting of cycle. For example, “heterocyclic group” includes compound below.
- (22) ##STR00002##
- (23) Unless otherwise stated, the term “aliphatic”, as used herein, means an aliphatic hydrocarbon having 1 to 60 carbon atoms, and the term “aliphatic ring”, as used herein, means an aliphatic hydrocarbon ring having 3 to 60 carbon atoms.
- (24) Unless otherwise stated, the term “ring”, as used herein, means an aliphatic ring having 3 to 60 carbon atoms, or an aromatic ring having 6 to 60 carbon atoms, or a hetero ring having 2 to 60 carbon atoms, or a fused ring formed by the combination of them, and includes a saturated or unsaturated ring.

(25) Other hetero compounds or hetero radicals other than the above-mentioned hetero compounds include, but are not limited thereto, one or more heteroatoms.

(26) Unless otherwise stated, the term “carbonyl”, as used herein, is represented by —COR', wherein R' may be hydrogen, an alkyl having 1 to 20 carbon atoms, an aryl having 6 to 30 carbon atoms, a cycloalkyl having 3 to 30 carbon atoms, an alkenyl having 2 to 20 carbon atoms, an alkynyl having 2 to 20 carbon atoms, or the combination of these.

(27) Unless otherwise stated, the term “ether”, as used herein, is represented by —R—O—R', wherein R or R' may be independently hydrogen, an alkyl having 1 to 20 carbon atoms, an aryl having 6 to 30 carbon atoms, a cycloalkyl having 3 to 30 carbon atoms, an alkenyl having 2 to 20 carbon atoms, an alkynyl having 2 to 20 carbon atoms, or the combination of these.

(28) Unless otherwise stated, the term “substituted or unsubstituted”, as used herein, means that substitution is substituted by at least one substituent selected from the group consisting of, but is not limited thereto, deuterium, halogen, an amino group, a nitrile group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.1-C.sub.20 alkylamine group, a C.sub.1-C.sub.20 alkylthio group, a C.sub.6-C.sub.20 arylthio group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.3-C.sub.20 cycloalkyl group, a C.sub.6-C.sub.20 aryl group, a C.sub.6-C.sub.20 aryl group substituted by deuterium, a C.sub.8-C.sub.20 arylalkenyl group, a silane group, a boron group, a germanium group, and a C.sub.2-C.sub.20 heterocyclic group.

(29) Unless otherwise expressly stated, the Formula used in the present invention, as used herein, is applied in the same manner as the substituent definition according to the definition of the exponent of the following Formula.

(30) ##STR00003##

(31) Here, when a is an integer of 0, it means that the substituent R^{sup.1} is absent, that is, when a is 0, it means that all hydrogens are bonded to carbons forming the benzene ring, and in this case, the display of hydrogen bonded to carbon may be omitted and the chemical formula or compound may be described.

(32) When a is an integer of 1, one substituent R^{sup.1} is bonded to any one of carbons forming a benzene ring, and when a is an integer of 2 or 3, they are respectively combined as follows, in which R^{sup.1} may be the same as or different from each other, and when a is an integer of 4 to 6, and it is bonded to the carbon of the benzene ring in a similar manner, whereas the indication of hydrogen bonded to the carbon forming the benzene ring is omitted.

(33) ##STR00004##

(34) Hereinafter, a compound according to an aspect of the present invention and an organic electric element comprising the same will be described.

(35) The present invention provides an organic electronic element comprising a first electrode, a second electrode, and an organic material layer formed between the first electrode and the second electrode, wherein the organic material layer comprises an emitting layer, wherein the emitting layer comprises a first host compound represented by Formula 1 and a second host compound represented by Formula 2 as the phosphorescent emitting layer.

(36) ##STR00005##

(37) {In Formula 1 and Formula 2, 1) A and B rings are each independently C.sub.6-C.sub.20 aryl or C.sub.2-C.sub.20 heterocycle; and, 2) X^{sup.1} is S or O, 3) X^{sup.2} is N-L^{sup.7}-Ar^{sup.9}, O, S or CR'R'',

(38) R' and R'' are each independently hydrogen; a C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.3-C.sub.60 heterocyclic group; a C.sub.1-C.sub.50 alkyl group; and -L'-N(R^{sup.a})

(R^{sup.b}), R' and R'' may be bonded to each other to form a spiro, 4) p and q are each independently an integer of 0~10, r is an integer of 0~3, s is an integer of 0~4, 5) R^{sup.1}, R^{sup.2}, R^{sup.3} and R^{sup.4} are each independently selected from a group consisting of hydrogen; a C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60 heterocyclic group including at

least one hetero atom of O, N, S, Si or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.1-C.sub.50 alkyl group; a C.sub.2-C.sub.20 alkenyl group; a C.sub.2-C.sub.20 alkynyl group; a C.sub.1-C.sub.30 alkoxyl group; a C.sub.6-C.sub.30 aryloxy group; and -L'-N(R.sup.a)(R.sup.b), and R.sup.a and R.sup.b are each independently selected from a group consisting of a C.sub.6-C.sub.60 aryl group; a fluorenyl group; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.2-C.sub.60 heterocyclic group including at least one hetero atom of O, N, S, Si or P; 6) wherein, L' is selected from the group consisting of a single bond; a C.sub.6-C.sub.60 arylene group; a fluorenylene group; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; and a C.sub.2-C.sub.60 heterocyclic, 7) Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, Ar.sup.7, Ar.sup.8 and Ar.sup.9 are each independently selected from the group consisting of a C.sub.6-C.sub.60 aryl group; a C.sub.2-C.sub.60 heterocyclic group including at least one hetero atom of O, N, S, Si or P; a fluorenyl group; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.1-C.sub.50 alkyl group; a C.sub.2-C.sub.20 alkenyl group; a C.sub.2-C.sub.20 alkynyl group; a C.sub.1-C.sub.30 alkoxyl group; C.sub.6-C.sub.30 arylthio group; a C.sub.6-C.sub.30 aryloxy group; and Ar.sup.1 and Ar.sup.2, Ar.sup.3 and Ar.sup.4, and Ar.sup.5 and Ar.sup.6 may be bonded to each other to form a ring, 8) L.sup.1, L.sup.2, L.sup.3, L.sup.4, L.sup.5, L.sup.5, L.sup.6 and L.sup.7 are independently selected from the group consisting of a single bond; a C.sub.6-C.sub.60 arylene group; a fluorenylene group; or a C.sub.2-C.sub.60 heteroarylene group containing at least one hetero atom of O, N, S, Si or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; and an aliphatic hydrocarbon group; 9) wherein, the aryl group, fluorenyl group, arylene group, heterocyclic group, fluorenylene group, fused ring group, alkyl group, alkenyl group, alkoxy group and aryloxy group may be substituted with one or more substituents selected from the group consisting of deuterium; halogen; a silane group; siloxane group; boron group; germanium group; cyano group; nitro group; a C.sub.1-C.sub.20 alkylthio group; C.sub.1-C.sub.20 alkoxyl group; C.sub.1-C.sub.20 alkyl group; C.sub.2-C.sub.20 alkenyl group; C.sub.2-C.sub.20 alkynyl group; C.sub.6-C.sub.20 aryl group; C.sub.6-C.sub.20 aryl group substituted with deuterium; a fluorenyl group; C.sub.2-C.sub.20 heterocyclic group; C.sub.3-C.sub.20 cycloalkyl group; C.sub.7-C.sub.20 arylalkyl group and C.sub.8-C.sub.20 arylalkenyl group, wherein the substituents may be bonded to each other to form a saturated or unsaturated ring, wherein the term 'ring' means a C.sub.3-C.sub.60 aliphatic ring or a C.sub.6-C.sub.60 aromatic ring or a C.sub.2-C.sub.60 heterocyclic group or a fused ring formed by the combination thereof.}

(39) Also, the present invention provides an organic electric element including a compound wherein A or B in Formula 1 is each independently selected from the group consisting of Formulas a-1 to a-7.

(40) ##STR00006##

(41) {In Formulas a-1 to a-7, Z.sup.1 to Z.sup.48 are each independently CR.sup.c or N, Z.sup.1 to Z.sup.48 bonded to L.sup.1 to L.sup.7 are carbon (C), R.sup.c is the same as the definition of R.sup.a, * indicates the position to be condensed.}

(42) Also, the present invention provides an organic electric element including a compound wherein L.sup.1 to L.sup.7 in Formula 1 or Formula 2 are represented by any one of the following Formulas b-1 to b-13.

(43) ##STR00007## ##STR00008##

(44) {In Formulas b-1 to b-13, Y is N-L.sup.B-Ar.sup.10, O, S or CR'R'', L.sup.8 is as the definition of L.sup.1, Ar.sup.10 is as the definition of Ar.sup.1, R' and R'' are the same as defined above, a, c, d and e are each independently an integer of 0 to 4, and b is an integer of 0 to 6, f and g are each independently an integer of 0 to 3, h is an integer of 0 to 2, i is an integer of 0 or 1, R.sup.5, R.sup.6 and R.sup.7 are each independently hydrogen; deuterium; tritium; halogen; cyano group; nitro group; C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60

heterocyclic group including at least one heteroatom of O, N, S, Si or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.1-C.sub.50 alkyl group; a C.sub.2-C.sub.20 alkenyl group; a C.sub.2-C.sub.20 alkynyl group; a C.sub.1-C.sub.30 alkoxy group; a C.sub.6-C.sub.30 aryloxy group; and -L.sup.a-N(R.sup.d)(R.sup.e); or in case a, b, c, d, e, f and g are 2 or more, and h is 2 or more, R.sup.5, R.sup.6 and R.sup.7 are in plural being the same as or different, and a plurality of R.sup.5 or a plurality of R.sup.6 or a plurality of R.sup.7 or adjacent R.sup.5 and R.sup.6, or adjacent R.sup.6 and R.sup.7 may be bonded to each other to form an aromatic or a heteroaromatic ring, where, L.sup.a is selected from the group consisting of a single bond; a C.sub.6-C.sub.60 arylene group; a fluorenylene group; a C.sub.2-C.sub.60 heterocyclic group containing at least one hetero atom of O, N, S, Si, or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; and C.sub.3-C.sub.60 aliphatic hydrocarbon group; R.sup.d and R.sup.e are each independently selected from the group consisting of a C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60 heterocyclic group containing at least one hetero atom of O, N, S, Si, or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring, Z.sup.49, Z.sup.50, and Z.sup.51 are each independently CR.sup.g or N, at least one of Z.sup.49, Z.sup.50, and Z.sup.51 is N, R.sup.g is selected from the group consisting of hydrogen; deuterium; tritium; halogen; cyano group; nitro group; C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60 heterocyclic group including at least one heteroatom of O, N, S, Si or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.1-C.sub.50 alkyl group; a C.sub.2-C.sub.20 alkenyl group; a C.sub.2-C.sub.20 alkynyl group; a C.sub.1-C.sub.30 alkoxy group; a C.sub.6-C.sub.30 aryloxy group; and adjacent R.sup.5 and R.sup.g may be bonded to each other to form an aromatic or a heteroaromatic ring. }

(45) As another example, the present invention provides an organic electric element including a compound in which at least one of Ar.sup.1 to Ar.sup.6 is represented by Formula 1-2.

(46) ##STR00009##

(47) {In Formula 1-2, C and D are the same as the definition of A, X.sup.3 is N-L.sup.10-Ar.sup.11, O, S or CR'R'', L.sup.9 and L.sup.10 are the same as the definition of L.sup.1, Ar.sup.11 is as the definition of Ar.sup.1, R' and R'' are the same as defined above. }

(48) The first host compound represented by Formula 1 includes a compound represented by Formula 3 or Formula 4 below.

(49) ##STR00010##

(50) {In Formula 3 or Formula 4, Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, L.sup.1, L.sup.2, L.sup.3, R.sup.1, R.sup.2 are the same as defined above, p' is an integer from 0 to 3, and q' is an integer from 0 to 2. }

(51) More specifically, the present invention provides an organic electric element wherein the first host compound represented by Formula 1 includes compounds represented by Formulas 5 to 11.

(52) ##STR00011##

(53) {In Formulas 5 to 11, L.sup.1, L.sup.2, L.sup.3, Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, R.sup.1, R.sup.2, X.sup.1 are the same as defined above, p' is an integer from 0 to 3, and q' is an integer from 0 to 2, o is an integer from 0 to 4. }

(54) In the present invention, the first host compound represented by Formula 1 includes compounds represented by the following Formulas 12 to 21.

(55) ##STR00012##

(56) {In Formulas 12 to 21, Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, L.sup.1, L.sup.2, L.sup.3, R.sup.1, R.sup.2, p, q, X.sup.1, A and B are the same as defined above, p' is an integer from 0 to 3, and q' is an integer from 0 to 2. }

(57) The compound represented by Formula 1 of the present invention includes Compound 1-1 to Compound 1-146 below.

(58) ##STR00013## ##STR00014## ##STR00015## ##STR00016## ##STR00017##

##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023##
##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029##
##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035##
##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041##
##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047##
##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052##

(59) Also, The second host compound represented by Formula 2 is represented by any one of Formulas 22 to 25 below.

(60) ##STR00053##

(61) {In Formulas 22 to 25, X^{sup.2}, L^{sup.4}, L^{sup.5}, L^{sup.6}, Ar^{sup.7}, Ar^{sup.8}, R^{sup.3}, R^{sup.4}, r and s are the same as defined above.}

(62) Specifically, in the present invention, the second host compound represented by Formula 2 includes a compound represented by Formula 26 below.

(63) ##STR00054##

(64) {In Formula 26, X^{sup.2}, L^{sup.4}, L^{sup.5}, L^{sup.6}, Ar^{sup.7}, R^{sup.3}, R^{sup.4}, r and s are the same as defined above, X^{sup.4} is the same as the definition of X^{sup.2}, R^{sup.8} and R^{sup.9} are the same as the definition of R^{sup.3} and R^{sup.4}, u is the same as the definition of r, and t is the same as the definition of s.}

(65) Also, in the present invention, the second host compound represented by Formula 2 includes a compound represented by Formulas 27 to 30 below.

(66) ##STR00055##

(67) {In Formulas 27 to 30, X^{sup.2}, L^{sup.4}, L^{sup.5}, L^{sup.6}, Ar^{sup.7}, R^{sup.3}, R^{sup.4}, r and s are the same as defined above, X^{sup.4}, R^{sup.8} and R^{sup.9} are the same as defined above.}

(68) In the present invention, the second host compound represented by Formula 2 includes the following compounds.

(69) ##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060##
##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066##
##STR00067## ##STR00068## ##STR00069## ##STR00070## ##STR00071## ##STR00072##
##STR00073## ##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078##
##STR00079## ##STR00080## ##STR00081## ##STR00082## ##STR00083## ##STR00084##
##STR00085## ##STR00086## ##STR00087## ##STR00088## ##STR00089##

(70) ##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094##
##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099## ##STR00100##
##STR00101## ##STR00102## ##STR00103## ##STR00104## ##STR00105## ##STR00106##
##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111## ##STR00112##
##STR00113## ##STR00114## ##STR00115## ##STR00116## ##STR00117## ##STR00118##
##STR00119## ##STR00120## ##STR00121##

(71) Referring to the FIGURE, the organic electric element(100) according to the present invention includes a first electrode(120) formed on a substrate(110), a second electrode(180), and an organic material layer including the compound represented by Formula 1 between the first electrode(120) and the second electrode(180). Here, the first electrode(120) may be an anode (positive electrode), and the second electrode(180) may be a cathode (negative electrode). In the case of an inverted organic electric element, the first electrode may be a cathode, and the second electrode may be an anode.

(72) The organic material layer may include a hole injection layer(130), a hole transport layer(140), an emitting layer(150), an emitting-auxiliary layer(151), an electron transport layer(160), and an electron injection layer(170) formed in sequence on the first electrode(120). Here, the remaining layers except the emitting layer(150) may not be formed. The organic material layer may further include a hole blocking layer, an electron blocking layer, an emitting-auxiliary layer(151), an electron transport auxiliary layer, a buffer layer(141), etc., and the electron transport layer(160) and

the like may serve as a hole blocking layer.

(73) Although not shown, the organic electric element according to the present invention may further include a protective layer formed on at least one side of the first and second electrodes, which is a side opposite to the organic material layer.

(74) Otherwise, even if the same core is used, the band gap, the electrical characteristics, the interface characteristics, and the like may vary depending on which substituent is bonded at which position, therefore the choice of core and the combination of sub-substituents associated therewith is also very important, and in particular, when the optimal combination of energy levels and T1 values and unique properties of materials (mobility, interfacial characteristics, etc.) of each organic material layer is achieved, a long lifespan and high efficiency can be achieved at the same time.

(75) The organic electroluminescent device according to an embodiment of the present invention may be manufactured using a PVD (physical vapor deposition) method. For example, an anode is formed by depositing a metal or a conductive metal oxide or an alloy thereof on a substrate, and after forming an organic material layer including the hole injection layer(130), the hole transport layer(140), the emitting layer(150), the electron transport layer(160) and the electron injection layer(170) thereon, the organic electroluminescent device according to an embodiment of the present invention can be manufactured by depositing a material that can be used as a cathode thereon.

(76) In addition, an emission auxiliary layer(151) may be further formed between the hole transport layer(140) and the emitting layer(150), and an electron transport auxiliary layer may be further formed between the emitting layer(150) and the electron transport layer (160).

(77) Accordingly, the present invention includes at least one hole transport layer between the first electrode and the emitting layer, wherein the hole transport layer includes a hole transport layer, an emitting auxiliary layer, or both, and wherein the hole transport layer includes the compound represented by Formula 1.

(78) Also, the compounds represented by Formula 1 and by Formula 2 are mixed in a ratio of any one of 1:9 to 9:1 to be included in the emitting layer, preferably mixed in a ratio of 1:9 to 5:5, more preferably in a ratio of 2:8 or 3:7 to be included in the emitting layer.

(79) The present invention may further include a light efficiency enhancing layer formed on at least one of the opposite side to the organic material layer among one side of the first electrode, or one of the opposite side to the organic material layer among one side of the second electrode.

(80) Also, the organic material layer is formed by one of a spin coating process, a nozzle printing process, an inkjet printing process, a slot coating process, a dip coating process or a roll-to-roll process, and since the organic material layer according to the present invention can be formed by various methods, the scope of the present invention is not limited by the method of forming the organic material layer.

(81) The organic electric element according to an embodiment of the present invention may be a front emission type, a back emission type, or a both-sided emission type, depending on the material used.

(82) WOLED (White Organic Light Emitting Device) is easy to realize high resolution and excellent processability, while there is an advantage that can be manufactured using the existing LCD color filter technology. Various structures for a white organic light emitting device mainly used as a backlight device have been proposed and patented. Typically, R(Red), G(Green), B(Blue) light emitting parts are arranged in a side-by-side manner, and R, G, B light emitting layers are stacked up and down, and blue (B) electroluminescence by organic emitting layer and, there is a color conversion material (CCM) method using photo-luminescence of an inorganic phosphor using light from this, and the present invention may be applied to such WOLED.

(83) The present invention also provides an electronic device comprising a display device including the organic electric element; and a control unit for driving the display device.

(84) According to another aspect, the present invention provides an electronic device wherein the

organic electric element is at least one of an OLED, an organic solar cell, an organic photo conductor, an organic transistor and an element for monochromic or white illumination. At this time, the electronic device may be a current or future wired/wireless communication terminal, and covers all kinds of electronic devices including a mobile communication terminal such as a cellular phone, a personal digital assistant (PDA), an electronic dictionary, a point-to-multipoint (PMP), a remote controller, a navigation unit, a game player, various kinds of TVs, and various kinds of computers.

(85) Hereinafter, Synthesis Examples of the compound represented by Formula 1 and Formula 2 according to the present invention and preparation examples of the organic electric element of the present invention will be described in detail by way of example, but are not limited to the following examples of the invention.

Synthesis Example 1

Synthesis of Formula 1

(86) The compound represented by Formula 1 according to the present invention (final product 1) is synthesized by reacting Sub 1 and Sub 2 as shown in Scheme 1 below, but is not limited thereto.

(87) ##STR00122##

I. Synthesis of Sub 1

(88) Sub 1 of Reaction Scheme 1 may be synthesized by Reaction Scheme 2 below, but is not limited thereto. (Hal.sup.1 is Br and Cl, and Hal.sup.2 is selected from I and Br.)

(89) ##STR00123##

(90) Also, Sub 1-I of Reaction Scheme 2 may be synthesized by Reaction Scheme 3 below, but is not limited thereto.

(91) ##STR00124##

(92) Synthesis examples of specific compounds belonging to Sub 1 are as follows.

1. Synthesis Example of Sub 1-1

(1) Synthesis of Sub 1-I-1

(93) ##STR00125##

[1] Synthesis of Sub 1-I-c-1

(94) To (4-Bromo-2-(methylthio)phenyl)boronic acid (20 g, 246.91 mmol), 3-bromo-5-iodophenol (24.2 g, 81.0 mmol), Pd(PPh.sub.3).sub.4 (2.81 g, 2.43 mmol), NaOH (6.48 g, 162 mmol), THF (200 mL), H.sub.2O (100 mL) were added and refluxed at 90° C. for 12 hours. When the reaction was completed, the temperature of the reaction product was cooled to room temperature, and extracted with CH.sub.2Cl.sub.2 and water, and the organic layer was dried over MgSO.sub.4 and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain the product. (25.5 g, Yield: 84%)

[2] Synthesis of Sub 1-I-d-1

(95) Add acetic acid (250 mL) to Sub1-I-c-1 (25.5 g, 68.1 mmol), add 35% Hydrogen peroxide (H.sub.2O.sub.2) (6.94 g), and stir at room temperature. When the reaction was completed, it was neutralized with an aqueous NaOH solution, followed by extraction with ethylacetate (EA) and water. The organic layer was dried over MgSO.sub.4 and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain the product. (21.9 g, Yield: 82%)

[3] Synthesis of Sub 1-I-e-1

(96) Sulfuric acid (H.sub.2SO.sub.4) (11 mL) was added to Sub1-I-d-1 (21.9 g, 56.0 mmol) and stirred at room temperature. When the reaction was completed, it was neutralized with an aqueous NaOH solution, and extracted with CH.sub.2Cl.sub.2 and water. The organic layer was dried over MgSO.sub.4 and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain the product. (16.8 g, Yield: 84%)

[4] Synthesis of Sub 1-I-1

(97) Sub1-Ie-1 (16.8 g, 46.8 mmol) was added to an excess of trifluoromethane-sulfonic acid, stirred at room temperature for 24 hours, then water and pyridine (8:1) were slowly added and

refluxed for 30 minutes. The temperature was lowered and extracted with CH₂Cl₂ and water. The organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain the product. (15.7 g, Yield: 68%)

(2) Synthesis of Sub 1-1

(98) ##STR00126##

[1] Synthesis of Sub 1-1

(99) After dissolving Sub1-I-1 (15.7 g, 32.0 mmol) with Toluene (210 mL), diphenylamine (10.8 g, 64.1 mmol), Pd₂(dba)₃ (1.76 g, 1.92 mmol), P(t-Bu)₃ (26.0 g, 64.0 mmol), NaOt-Bu (12.3 g, 128 mmol) were added and stirred at 90° C. When the reaction was completed, the temperature of the reaction product was cooled to room temperature, and extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain Sub 1-1. (13.7 g, Yield: 64%)

2. Synthesis Example of Sub 1-3

(100) ##STR00127##

[1] Synthesis of Sub 1-I-h-3

(101) (4-bromo-2-hydroxyphenyl)boronic acid (20 g, 92.2 mmol) on 3-bromo-5-iodophenol (27.6 g, 92.2 mmol), Pd(PPh₃)₄ (3.20 g, 2.77 mmol), NaOH (7.38 g, 184.5 mmol), THF (200 mL), H₂O (100 mL) were added and refluxed at 90° C. for 12 hours. When the reaction was completed, the temperature of the reaction product was cooled to room temperature, and extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain the product. (24.0 g, Yield: 76%)

[2] Synthesis of Sub 1-I-i-3

(102) Sub1-I-h-3 (24.0 g, 69.8 mmol) on Pd(OAc)₂ (0.78 g, 3.49 mmol), 3-nitropyridine (0.43 g, 3.49 mmol), BzOOt-Bu (tert-butyl peroxybenzoate) (27.1 g, 139.5 mmol), C₆F₆ (hexafluorobenzene) (160 mL), DMI (N,N-dimethylimidazolidinone) (100 mL) were added and refluxed at 90° C. for 12 hours. When the reaction was completed, the temperature of the reaction product was cooled to room temperature, and extracted with EA and water. The organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain the product. (14.3 g, Yield: 60%)

[3] Synthesis of Sub 1-I-3

(103) Sub1-I-i-3 (14.3 g, 41.7 mmol) was added to an excess of trifluoromethane-sulfonic acid, stirred at room temperature for 24 hours, then water and pyridine (8:1) were slowly added and refluxed for 30 minutes. The temperature was lowered and extracted with CH₂Cl₂ and water. The organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain the product. (15.4 g, Yield: 78%)

[4] Synthesis of Sub 1-3

(104) After dissolving Sub1-I-3 (15.4 g, 32.4 mmol) with Toluene (200 mL), diphenylamine (11.0 g, 64.8 mmol), Pd₂(dba)₃ (1.78 g, 1.94 mmol), P(t-Bu)₃ (26.4 g, 64.8 mmol), NaOt-Bu (12.5 g, 130 mmol) were added and stirred at 90° C. When the reaction was completed, the temperature of the reaction product was cooled to room temperature, and extracted with CH₂Cl₂ and water. The organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain Sub 1-3. (14.4 g, Yield: 68%)

3. Synthesis Example of Sub 1-30

(105) ##STR00128## ##STR00129##

[1] Synthesis of Sub 1-I-c-30

(106) (5-Bromo-2-(methylthio)phenyl)boronic acid (20 g, 81.0 mmol), 3-bromo-5-iodophenol (24.2 g, 81.0 mmol), Pd(PPh₃)₄ (2.81 g, 2.43 mmol), NaOH (6.48 g, 2.43 mmol) were

used to obtain a product (24.4 g, 81%) using the synthesis method of Sub 1-I-c-1.

[2] Synthesis of Sub 1-I-d-30

(107) Sub 1-I-c-30 (24.4 g, 65.3 mmol), acetic acid (245 mL), 35% Hydrogen peroxide (H.sub.2O.sub.2) (6.66 g) were used to obtain a product (20.1 g, 79%) using the synthesis method of Sub 1-I-d-1.

[3] Synthesis of Sub 1-I-e-30

(108) Sub 1-I-d-30 (20.1 g, 51.6 mmol), Sulfuric acid (H.sub.2SO.sub.4) (10 mL) were used to obtain a product (15.5 g, 84%) using the synthesis method of Sub 1-I-e-1.

[4] Synthesis of Sub 1-I-30

(109) Sub 1-I-e-30 (15.5 g, 42.3 mmol) was added to an excess of trifluoromethane-sulfonic acid, and were used to obtain a product (16 g, 75%) using the synthesis method of Sub 1-I-1.

[5] Synthesis of Sub 1-30

(110) Sub1-I-30 (15.5 g, 32.6 mmol), diphenylamine (11.0 g, 65.2 mmol), Pd.sub.2(dba).sub.3 (1.79 g, 2.00 mmol), P(t-Bu).sub.3 (26.4 g, 65.2 mmol), NaOt-Bu (12.5 g, 130 mmol) were used to obtain Sub 1-30 (14.0 g, 64.5%) using the synthesis method of Sub 1-1.

4. Synthesis Example of Sub 1-33

(111) ##STR00130## ##STR00131##

[1] Synthesis of Sub 1-I-h-33

(112) (5-chloro-2-hydroxyphenyl)boronic acid (20 g, 116 mmol), 3-bromo-4-iodophenol (34.8 g, 116 mmol), Pd(PPh.sub.3).sub.4 (4.03 g, 3.49 mmol), NaOH (9.30 g, 233 mmol) were used to obtain a product (27.2 g, 78%) using the synthesis method of Sub 1-I-h-3.

[2] Synthesis of Sub 1-I-i-33

(113) Sub1-I-h-33 (27.2 g, 90.7 mmol), Pd(OAc).sub.2 (1.02 g, 4.54 mmol), 3-nitropyridine (0.56 g, 4.54 mmol), BzOOt-Bu (tert-butyl peroxybenzoate) (35.2 g, 181 mmol), C.sub.6F.sub.6 (hexafluorobenzene) (210 mL), DMI (N,N'-dimethylimidazolidinone) (135 mL) were used to obtain a product (14.7 g, 54%) using the synthesis method of Sub 1-I-i-3.

[3] Synthesis of Sub 1-I-33

(114) Sub1-I-i-33 (14.7 g, 49.3 mmol) was added to an excess of trifluoromethane-sulfonic acid, and were used to obtain a product (14.7 g, 69%) using the synthesis method of Sub 1-I-3.

[4] Synthesis of Sub 1-II-33

(115) Sub1-I-33 (14.7 g, 34.1 mmol), diphenylamine (5.77 g, 34.1 mmol), Pd.sub.2(dba).sub.3 (0.94 g, 1.02 mmol), P(t-Bu).sub.3 (6.83 g, 34.1 mmol), NaOt-Bu (6.56 g, 68.2 mmol) were used to obtain Sub 1-II-33 (12.3 g, 70%) using the synthesis method of Sub 1-1.

[5] Synthesis of Sub 1-33

(116) Sub1-II-33 (12.3 g, 23.7 mmol), N-phenyldibenzo[b,d]thiophen-3-amine (6.54 g, 23.7 mmol), Pd.sub.2(dba).sub.3 (0.65 g, 0.71 mmol), P(t-Bu).sub.3 (4.83 g, 23.7 mmol), NaOt-Bu (4.56 g, 47.5 mmol) were used to obtain Sub 1-33 (12.3 g, 70%) using the synthesis method of Sub 1-1.

5. Synthesis Example of Sub 1-46

(117) ##STR00132##

[1] Synthesis of Sub 1-I-c-46

(118) (5-chloro-2-(methylthio)phenyl)boronic acid (20 g, 98.8 mmol), 4-bromo-3-iodophenol (29.53 g, 98.8 mmol), Pd(PPh.sub.3).sub.4 (3.42 g, 2.96 mmol), NaOH (7.90 g, 198 mmol) were used to obtain a product (27.2 g, 84%) using the synthesis method of Sub 1-I-c-1.

[2] Synthesis of Sub 1-I-d-46

(119) Sub 1-I-c-46 (27.2 g, 82.6 mmol), acetic acid (270 mL), 35% Hydrogen peroxide (H.sub.2O.sub.2) (8.43 g) were used to obtain a product (21.8 g, 76%) using the synthesis method of Sub 1-I-d-1.

[3] Synthesis of Sub 1-I-e-4

(120) Sub 1-I-d-46 (21.8 g, 62.9 mmol), Sulfuric acid (H.sub.2SO.sub.4) (11 mL) were used to

obtain a product (15.7 g, 79%) using the synthesis method of Sub 1-I-e-1.

[4] Synthesis of Sub 1-I-46

(121) Sub 1-I-e-46 (15.7 g, 50.0 mmol) was added to an excess of trifluoromethane-sulfonic acid, and were used to obtain a product (15.7 g, 71%) using the synthesis method of Sub 1-I-1.

[5] Synthesis of Sub 1-II-46

(122) Sub1-I-46 (15.7 g, 35.3 mmol), 9H-carbazole (5.91 g, 35.3 mmol), Pd.sub.2(dba).sub.3 (0.97 g, 1.06 mmol), P(t-Bu).sub.3 (7.17 g, 35.3 mmol), NaOt-Bu (6.79 g, 70.7 mmol) were used to obtain Sub 1-II-46 (11.7 g, 62%) using the synthesis method of Sub 1-1.

[6] Synthesis of Sub 1-46

(123) Sub1-II-46 (11.7 g, 22.0 mmol), diphenylamine (3.71 g, 22.0 mmol), Pd.sub.2(dba).sub.3 (0.60 g, 0.66 mmol), P(t-Bu).sub.3 (4.50 g, 22.0 mmol), NaOt-Bu (4.22 g, 43.9 mmol) were used to obtain Sub 1-46 (9.77 g, 67%) using the synthesis method of Sub 1-1.

6. Synthesis Example of Sub 1-53

(124) ##STR00133##

[1] Synthesis of Sub 1-I-h-53

(125) (2-bromo-6-hydroxy-4-methylphenyl)boronic acid (20 g, 86.6 mmol) on 3-bromo-5-iodophenol (25.9 g, 86.6 mmol), Pd(PPh.sub.3).sub.4 (3.00 g, 2.60 mmol), NaOH (6.93 g, 173 mmol) were used to obtain a product (26 g, 84%) using the synthesis method of Sub 1-I-h-3.

[2] Synthesis of Sub 1-I-i-53

(126) Sub 1-I-h-53 (26.0 g, 72.6 mmol) on Pd(OAc).sub.2 (0.82 g, 3.63 mmol), 3-nitropyridine (0.45 g, 3.63 mmol), BzOOt-Bu (tert-butyl peroxybenzoate) (28.2 g, 145 mmol), C.sub.6F.sub.6 (hexafluorobenzene) (170 mL), DMI (N,N'-dimethylimidazolidinone) (110 mL) were used to obtain a product (13.3 g, 51%) using the synthesis method of Sub 1-I-i-3.

[3] Synthesis of Sub 1-I-53

(127) Sub 1-I-i-53 (13.3 g, 37.3 mmol) was added to an excess of trifluoromethane-sulfonic acid, and were used to obtain a product (13.2 g, 72%) using the synthesis method of Sub 1-I-3.

[4] Synthesis of Sub 1-I-53

(128) Sub1-I-53 (13.2 g, 27.0 mmol), diphenylamine (9.14 g, 54.0 mmol), Pd.sub.2(dba).sub.3 (1.48 g, 1.62 mmol), P(t-Bu).sub.3 (22.0 g, 54.0 mmol), NaOt-Bu (10.4 g, 108 mmol) were used to obtain Sub 1-53 (11.0 g, 61%) using the synthesis method of Sub 1-1.

7. Synthesis Example of Sub 1-81

(129) ##STR00134## ##STR00135##

[1] Synthesis of Sub 1-I-c-81

(130) (3-bromo-2-(methylthio)phenyl)boronic acid (20 g, 81.0 mmol), 4-bromo-3-iodonaphthalen-1-ol (28.3 g, 81.0 mmol), Pd(PPh.sub.3).sub.4 (2.81 g, 2.43 mmol), NaOH (6.48 g, 162 mmol) were used to obtain a product (29.0 g, 84%) using the synthesis method of Sub 1-I-c-1.

[2] Synthesis of Sub 1-I-d-81

(131) Sub1-I-c-81 (29.0 g, 68.3 mmol), acetic acid (290 mL), 35% Hydrogen peroxide (H.sub.2O.sub.2) (6.96 g) were used to obtain a product (24.4 g, 81%) using the synthesis method of Sub 1-I-d-1.

[3] Synthesis of Sub 1-I-e-81

(132) Sub1-I-d-81 (24.4 g, 55.3 mmol), Sulfuric acid (H.sub.2SO.sub.4) (12 mL) were used to obtain a product (18.0 g, 80%) using the synthesis method of Sub 1-I-e-1.

[4] Synthesis of Sub 1-I-81

(133) Sub1-I-e-81 (18.0 g, 44.1 mmol) was added to an excess of trifluoromethane-sulfonic acid, and were used to obtain a product (17.0 g, 71%) using the synthesis method of Sub 1-I-1.

[5] Synthesis of Sub 1-81

(134) Sub1-I-81 (17.0 g, 31.5 mmol), diphenylamine (10.7 g, 63.1 mmol), Pd.sub.2(dba).sub.3 (1.73 g, 1.89 mmol), P(t-Bu).sub.3 (25.6 g, 63 mmol), NaOt-Bu (12.1 g, 126 mmol) were used to obtain Sub 1-81 (15.3 g, 68%) using the synthesis method of Sub 1-1.

8. Synthesis Example of Sub 1-109

(135) ##STR00136##

[1] Synthesis of Sub1-I-h-109

(136) (3-bromo-2-hydroxynaphthalen-1-yl)boronic acid (g, mmol) on 3-bromo-5-iodophenol (20 g, 74.9 mmol), Pd(PPh.sub.3).sub.4 (22.4 g, 74.9 mmol), NaOH (6.00 g, 150 mmol) were used to obtain a product (23.1 g, 78%) using the synthesis method of Sub 1-I-h-1.

[2] Synthesis of Sub 1-I-i-109

(137) Sub1-I-h-109 (23.1 g, 58.5 mmol) on Pd(OAc).sub.2 (0.66 g, 2.93 mmol), 3-nitropyridine (0.36 g, 2.93 mmol), BzOOt-Bu (tert-butyl peroxybenzoate) (22.7 g, 117 mmol), C.sub.6F.sub.6 (hexafluorobenzene) (135 mL), DMI (N,N'-dimethylimidazolidinone) (90 mL) were used to obtain a product (12.5 g, 54.5%) using the synthesis method of Sub 1-I-i-3.

[3] Synthesis of Sub 1-I-109

(138) Sub1-I-i-109 (12.5 g, 31.9 mmol) was added to an excess of trifluoromethane-sulfonic acid, and were used to obtain a product (11.7 g, 22.3%) using the synthesis method of Sub 1-I-3.

[4] Synthesis of Sub 1-109

(139) Sub1-I-109 (11.7 g, 22.3 mmol), diphenylamine (7.55 g, 44.6 mmol), Pd.sub.2(dba).sub.3 (1.22 g, 1.34 mmol), P(t-Bu).sub.3 (18.0 g, 44.6 mmol), NaOt-Bu (8.57 g, 89.2 mmol) were used to obtain Sub 1-109 (10.2 g, 65%) using the synthesis method of Sub 1-1.

9. Synthesis Example of Sub 1-133

(140) ##STR00137##

[1] Synthesis of Sub 1-II-133

(141) Sub 1-I-20 (15 g, 35.1 mmol), N-phenylnaphthalen-1-amine (7.69 g, 35.1 mmol), Pd.sub.2(dba).sub.3 (0.96 g, 1.05 mmol), P(t-Bu).sub.3 (7.17 g, 35.1 mmol), NaOt-Bu (6.74 g, 70.2 mmol) were used to obtain Sub 1-II-133 (12.5 g, 62%) using the synthesis method of Sub 1-1.

[2] Synthesis of Sub 1-133

(142) Sub1-II-133 (12.5 g, 22.0 mmol), N-phenylquinolin-7-amine (4.85 g, 22.0 mmol), Pd.sub.2(dba).sub.3 (0.60 g, 0.66 mmol), P(t-Bu).sub.3 (4.50 g, 22.0 mmol), NaOt-Bu (4.23 g, 44.0 mmol) were used to obtain Sub 1-133 (10.9 g, 66%) using the synthesis method of Sub 1-1.

10. Synthesis Example of Sub 1-137

(143) ##STR00138##

[1] Synthesis of Sub 1-I-c-137

(144) (3-bromo-2-(methylthio)phenyl)boronic acid (20 g, 71.8 mmol), 4-bromo-3-iodonaphthalen-1-ol (41.9 g, 71.8 mmol), Pd(PPh.sub.3).sub.4 (2.49 g, 2.15 mmol), NaOH (5.74 g, 144 mmol) were used to obtain a product (38.9 g, 79%) using the synthesis method of Sub 1-I-I-c-1.

[2] Synthesis of Sub 1-I-d-137

(145) Sub1-I-c-137 (38.9 g, 56.4 mmol), acetic acid (400 mL), 35% Hydrogen peroxide (H.sub.2O.sub.2) (5.75 g) were used to obtain a product (31.7 g, 80%) using the synthesis method of Sub 1-I-I-d-1.

[3] Synthesis of Sub 1-I-e-137

(146) Sub1-I-d-137 (31.7 g, 45.0 mmol), Sulfuric acid (H.sub.2SO.sub.4) (16 mL) were used to obtain a product (19.9 g, 82%) using the synthesis method of Sub 1-I-e-1.

[4] Synthesis of Sub 1-I-137

(147) Sub1-I-e-137 (19.9 g, 36.7 mmol) was added to an excess of trifluoromethane-sulfonic acid, and were used to obtain a product (17.4 g, 70%) using the synthesis method of Sub 1-I-1.

[5] Synthesis of Sub 1-II-137

(148) Sub1-I-137 (17.4 g, 25.8 mmol), Sub2-1 (4.36 g, 25.8 mmol), Pd.sub.2(dba).sub.3 (0.71 g, 0.77 mmol), P(t-Bu).sub.3 (5.17 g, 25.8 mmol), NaOt-Bu (4.96 g, 51.6 mmol) were used to obtain Sub 1-II-137 (12.8 g, 65%) using the synthesis method of Sub 1-1.

[6] Synthesis of Sub 1-137

(149) Sub1-II-137 (12.8 g, 16.8 mmol), Sub2-11 (3.69 g, 16.8 mmol), Pd.sub.2(dba).sub.3 (0.46 g,

0.51 mmol), P(t-Bu).sub.3 (3.33 g, 16.8 mmol), NaOt-Bu (3.24 g, 33.7 mmol) were used to obtain Sub 1-137 (10.3 g, 65%) using the synthesis method of Sub 1-1.

(150) The compound belonging to Sub 1 may be the following compound, but is not limited thereto, and Table 1 shows the FD-MS (Field Desorption-Mass Spectrometry) values of some compounds belonging to Sub 1.

(151) ##STR00139## ##STR00140## ##STR00141## ##STR00142## ##STR00143##
##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148## ##STR00149##
##STR00150## ##STR00151## ##STR00152## ##STR00153## ##STR00154## ##STR00155##
##STR00156##

(152) TABLE-US-00002 TABLE 1 compound FD-MS compound FD-MS Sub1-1 m/z =
666.13(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 666.73) Sub1-2 m/z =
756.14(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.4S.sub.2 = 756.81) Sub1-3 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-4 m/z =
831.18(C.sub.49H.sub.32F.sub.3N.sub.3O.sub.3S.sub.2 = 831.93) Sub1-5 m/z =
664.11(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.3S.sub.2 = 664.72) Sub1-6 m/z =
714.13(C.sub.41H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 714.78) Sub1-7 m/z =
742.16(C.sub.43H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 742.83) Sub1-8 m/z =
666.13(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 666.73) Sub1-9 m/z =
756.14(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.4S.sub.2 = 756.81) Sub1-10 m/z =
756.14(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.4S.sub.2 = 756.81) Sub1-11 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-12 m/z =
714.13(C.sub.41H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 714.78) Sub1-13 m/z =
648.13(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.4S = 648.66) Sub1-14 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-15 m/z =
666.13(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 666.73) Sub1-16 m/z =
815.21(C.sub.49H.sub.32F.sub.3N.sub.3O.sub.4S = 815.87) Sub1-17 m/z =
772.11(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.3S.sub.3 = 772.88) Sub1-18 m/z =
772.11(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.3S.sub.3 = 772.88) Sub1-19 m/z =
664.11(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.3S.sub.2 = 664.72) Sub1-20 m/z =
648.13(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.4S = 648.66) Sub1-21 m/z =
743.15(C.sub.42H.sub.28F.sub.3N.sub.3O.sub.3S.sub.2 = 743.82) Sub1-22 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-23 m/z =
802.21(C.sub.49H.sub.33F.sub.3N.sub.2O.sub.4S = 802.87) Sub1-24 m/z =
691.12(C.sub.38H.sub.24F.sub.3N.sub.3O.sub.3S.sub.2 = 691.74) Sub1-25 m/z =
667.12(C.sub.36H.sub.24F.sub.3N.sub.3O.sub.3S.sub.2 = 667.72) Sub1-26 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-27 m/z =
882.18(C.sub.53H.sub.33F.sub.3N.sub.2O.sub.4S.sub.2 = 882.97) Sub1-28 m/z =
772.11(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.3S.sub.3 = 772.88) Sub1-29 m/z =
740.16(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.5S = 740.75) Sub1-30 m/z =
666.13(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 666.73) Sub1-31 m/z =
726.18(C.sub.43H.sub.29F.sub.3N.sub.2O.sub.4S = 726.77) Sub1-32 m/z =
671.16(C.sub.37H.sub.20D.sub.5F.sub.3N.sub.2O.sub.3S.sub.2 = 671.76) Sub1-33 m/z =
756.14(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.4S.sub.2 = 756.81) Sub1-34 m/z =
904.20(C.sub.56H.sub.35F.sub.3N.sub.2O.sub.3S.sub.2 = 905.02) Sub1-35 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-36 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-37 m/z =
666.13(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 666.73) Sub1-38 m/z =
675.14(C.sub.38H.sub.24F.sub.3N.sub.3O.sub.4S = 675.68) Sub1-39 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-40 m/z =
772.11(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.3S.sub.3 = 772.88) Sub1-41 m/z =

782.19(C.sub.46H.sub.33F.sub.3N.sub.2O.sub.3S.sub.2 = 782.90) Sub1-43 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-43 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-44 m/z =
818.19(C.sub.49H.sub.33F.sub.3N.sub.2O.sub.3S.sub.2 = 818.93) Sub1-45 m/z =
648.13(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.4S = 648.66) Sub1-46 m/z =
664.11(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.3S.sub.2 = 664.72) Sub1-47 m/z =
651.14(C.sub.36H.sub.24F.sub.3N.sub.3O.sub.4S = 651.66) Sub1-48 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-49 m/z =
831.18(C.sub.49H.sub.32F.sub.3N.sub.3O.sub.3S.sub.2 = 831.93) Sub1-50 m/z =
740.16(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.5S = 740.75) Sub1-51 m/z =
772.11(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.3S.sub.3 = 772.88) Sub1-52 m/z =
742.16(C.sub.43H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 742.83) Sub1-53 m/z =
664.16(C.sub.38H.sub.27F.sub.3N.sub.2O.sub.4S = 664.70) Sub1-54 m/z =
648.13(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.4S = 648.66) Sub1-55 m/z =
664.11(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.3S.sub.2 = 664.72) Sub1-56 m/z =
696.14(C.sub.38H.sub.27F.sub.3N.sub.2O.sub.4S.sub.2 = 696.76) Sub1-57 m/z =
666.13(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 666.73) Sub1-58 m/z =
766.21(C.sub.46H.sub.33F.sub.3N.sub.2O.sub.4S = 766.84) Sub1-59 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-60 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-61 m/z =
664.11(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.3S.sub.2 = 664.72) Sub1-62 m/z =
664.11(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.3S.sub.2 = 664.72) Sub1-63 m/z =
690.18(C.sub.40H.sub.29F.sub.3N.sub.2O.sub.4S = 690.74) Sub1-64 m/z =
666.13(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.3S.sub.2 = 666.73) Sub1-65 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-66 m/z =
740.16(C.sub.43H.sub.27F.sub.3N.sub.2O.sub.5S = 740.75) Sub1-67 m/z =
907.22(C.sub.55H.sub.36F.sub.3N.sub.3O.sub.3S.sub.2 = 908.03) Sub1-68 m/z =
726.18(C.sub.43H.sub.29F.sub.3N.sub.2O.sub.4S = 726.77) Sub1-69 m/z =
664.11(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.3S.sub.2 = 664.72) Sub1-70 m/z =
648.13(C.sub.37H.sub.23F.sub.3N.sub.2O.sub.4S = 648.66) Sub1-71 m/z =
650.15(C.sub.37H.sub.25F.sub.3N.sub.2O.sub.4S = 650.67) Sub1-72 m/z =
833.20(C.sub.49H.sub.34F.sub.3N.sub.3O.sub.3S.sub.2 = 833.94) Sub1-73 m/z =
726.18(C.sub.43H.sub.29F.sub.3N.sub.2O.sub.4S = 726.77) Sub1-74 m/z =
818.19(C.sub.49H.sub.33F.sub.3N.sub.2O.sub.3S.sub.2 = 818.93) Sub1-75 m/z =
833.20(C.sub.49H.sub.34F.sub.3N.sub.3O.sub.3S.sub.2 = 833.94) Sub1-76 m/z =
833.20(C.sub.49H.sub.34F.sub.3N.sub.3O.sub.3S.sub.2 = 833.94) Sub1-77 m/z =
817.22(C.sub.49H.sub.34F.sub.3N.sub.3O.sub.4S = 817.88) Sub1-78 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-79 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-80 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-81 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-82 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-83 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-84 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-85 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-86 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-87 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-88 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-89 m/z =
716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-90 m/z =
700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-91 m/z =

700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-92 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-93 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-94 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-95 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-96 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-97 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-98 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-99 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-100 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-101 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-102 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-103 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-104 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-105 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-106 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-107 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-108 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-109 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-110 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O3S.sub.2 = 716.79) Sub1-111 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-112 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-113 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-114 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-115 m/z =
 716.14(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.3S.sub.2 = 716.79) Sub1-116 m/z =
 700.16(C.sub.41H.sub.27F.sub.3N.sub.2O.sub.4S = 700.73) Sub1-117 m/z =
 766.16(C.sub.45H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 766.85) Sub1-118 m/z =
 766.16(C.sub.45H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 766.85) Sub1-119 m/z =
 750.18(C.sub.45H.sub.29F.sub.3N.sub.2O.sub.4S = 750.79) Sub1-120 m/z =
 766.16(C.sub.45H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 766.85) Sub1-121 m/z =
 766.16(C.sub.45H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 766.85) Sub1-122 m/z =
 750.18(C.sub.45H.sub.29F.sub.3N.sub.2O.sub.4S = 750.79) Sub1-123 m/z =
 766.16(C.sub.45H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 766.85) Sub1-124 m/z =
 796.17(C.sub.46H.sub.31F.sub.3N.sub.2O.sub.4S.sub.2 = 796.88) Sub1-125 m/z =
 818.19(C.sub.49H.sub.33F.sub.3N.sub.2O.sub.3S.sub.2 = 818.93) Sub1-126 m/z =
 726.18(C.sub.43H.sub.29F.sub.3N.sub.2O.sub.4S = 726.77) Sub1-127 m/z =
 742.16(C.sub.43H.sub.29F.sub.3N.sub.2O.sub.3S.sub.2 = 742.83) Sub1-128 m/z =
 1058.25(C.sub.67H.sub.41F.sub.3N.sub.2O.sub.4S.sub.2 = 1059.19) Sub1-129 m/z =
 868.20(C.sub.53H.sub.35F.sub.3N.sub.2O.sub.3S.sub.2 = 868.99) Sub1-130 m/z =
 892.26(C.sub.56H.sub.39F.sub.3N.sub.2O.sub.4S = 892.99) Sub1-131 m/z =
 1112.32(C.sub.72H.sub.43D.sub.4F.sub.3N.sub.2O.sub.3S.sub.2 = 1113.32) Sub1-132 m/z =
 834.16(C.sub.47H.sub.29F.sub.3N.sub.4O.sub.4S.sub.2 = 834.89) Sub1-133 m/z =
 751.18(C.sub.44H.sub.28F.sub.3N.sub.3O.sub.4S = 751.78) Sub1-134 m/z =
 1135.28(C.sub.71H.sub.44F.sub.3N.sub.5O.sub.3S.sub.2 = 1136.28) Sub1-135 m/z =
 928.26(C.sub.59H.sub.39F.sub.3N.sub.2O.sub.4S = 929.03) Sub1-136 m/z =
 994.25(C.sub.63H.sub.41F.sub.3N.sub.2O.sub.3S.sub.2 = 995.15) Sub1-137 m/z =
 944.24(C.sub.59H.sub.39F.sub.3N.sub.2O.sub.3S.sub.2 = 945.09) Sub1-138 m/z =
 727.18(C.sub.42H.sub.28F.sub.3N.sub.3O.sub.4S = 727.76)

II. Synthesis of Sub 2

(153) Sub 2 of Reaction Scheme 1 may be synthesized (initiated in Korean Patent Registration No.

10-1251451 (registered on Apr. 5, 2013) of the applicant) by the reaction path of Scheme 4 below, but is not limited thereto. Z.sup.1 is Ar.sup.1 or Ar.sup.3, Z.sup.2 is Ar.sup.2 or Ar.sup.4.

(154) ##STR00157##

(155) Compounds belonging to Sub 2 may be the following compounds, but are not limited thereto, and Table 2 shows FD-MS (Field Desorption-Mass Spectrometry) values of some compounds belonging to Sub 2.

(156) ##STR00158## ##STR00159## ##STR00160## ##STR00161## ##STR00162##

(157) TABLE-US-00003 TABLE 2 compound FD-MS compound FD-MS Sub2-1 m/z = 169.09(C.sub.12H.sub.11N = 169.23) Sub2-2 m/z = 174.12(C.sub.12H.sub.6D.sub.5N = 174.26) Sub2-3 m/z = 245.12(C.sub.18H.sub.15N = 245.33) Sub2-4 m/z = 321.15(C.sub.24H.sub.19N = 321.42) Sub2-5 m/z = 209.12(C.sub.15H.sub.15N = 209.29) Sub2-6 m/z = 215.08(C.sub.13H.sub.13NS = 215.31) Sub2-7 m/z = 245.12(C.sub.18H.sub.15N = 245.33) Sub2-8 m/z = 194.08(C.sub.13H.sub.10N.sub.2 = 194.24) Sub2-9 m/z = 187.08(C.sub.12H.sub.10FN = 187.22) Sub2-10 m/z = 205.07(C.sub.12H.sub.9F.sub.2N = 205.21) Sub2-11 m/z = 219.10(C.sub.16H.sub.13N = 219.29) Sub2-12 m/z = 219.10(C.sub.16H.sub.13N = 219.29) Sub2-13 m/z = 269.12(C.sub.20H.sub.15N = 269.35) Sub2-14 m/z = 220.10(C.sub.15H.sub.12N.sub.2 = 200.28) Sub2-15 m/z = 170.08(C.sub.11H.sub.10N.sub.2 = 170.22) Sub2-16 m/z = 167.07(C.sub.12H.sub.9N = 167.21) Sub2-17 m/z = 217.09(C.sub.16H.sub.11N = 217.27) Sub2-18 m/z = 217.09(C.sub.16H.sub.11N = 217.27) Sub2-19 m/z = 217.09(C.sub.16H.sub.11N = 217.27) Sub2-20 m/z = 285.15(C.sub.21H.sub.19N = 285.39) Sub2-21 m/z = 285.15(C.sub.21H.sub.19N = 285.39) Sub2-22 m/z = 407.17(C.sub.31H.sub.21N = 407.52) Sub2-23 m/z = 485.21(C.sub.37H.sub.27N = 485.63) Sub2-24 m/z = 334.15(C.sub.24H.sub.18N.sub.2 = 334.42) Sub2-25 m/z = 334.15(C.sub.24H.sub.18N.sub.2 = 334.42) Sub2-26 m/z = 334.15(C.sub.24H.sub.18N.sub.2 = 334.42) Sub2-27 m/z = 334.15(C.sub.24H.sub.18N.sub.2 = 334.42) Sub2-28 m/z = 384.16(C.sub.28H.sub.20N.sub.2 = 384.48) Sub2-29 m/z = 436.17(C.sub.30H.sub.20N.sub.4 = 436.52) Sub2-30 m/z = 259.10(C.sub.18H.sub.13NO = 259.31) Sub2-31 m/z = 259.10(C.sub.18H.sub.13NO = 259.31) Sub2-32 m/z = 261.09(C.sub.16H.sub.11N.sub.3O = 261.28) Sub2-33 m/z = 275.08(C.sub.18H.sub.13NS = 275.37) Sub2-34 m/z = 275.08(C.sub.18H.sub.13NS = 275.37) Sub2-35 m/z = 275.08(C.sub.18H.sub.13NS = 275.37) Sub2-36 m/z = 275.08(C.sub.18H.sub.13NS = 275.37)

III. Synthesis of Final Product 1

(158) After dissolving Sub 1 (1 eq.) with Toluene in a round bottom flask, Sub 2 (1 eq.), Pd.sub.2(dba).sub.3 (0.03 eq.), (t-Bu).sub.3P (0.06 eq.), and NaOt-Bu (2 eq.) were stirred at 100° C. When the reaction was completed, the resulting compound was extracted with CH.sub.2Cl.sub.2 and water, and the organic layer was dried over MgSO.sub.4 and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain Final product 1.

1. Synthesis Example of 1-1

(159) ##STR00163##

(160) After dissolving Sub 1-1 (13.7 g, 20.5 mmol) with Toluene (180 mL) in a round bottom flask, Sub 2-1 (3.48 g, 20.5 mmol), Pd.sub.2(dba).sub.3 (0.56 g, 0.62 mmol), P(t-Bu).sub.3 (4.16 g, 20.5 mmol), NaOt-Bu (3.95 g, 41.1 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH.sub.2Cl.sub.2 and water, and the organic layer was dried over MgSO.sub.4 and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-1 (10.5 g, yield: 74%)

2. Synthesis Example of 1-3

(161) ##STR00164##

(162) After dissolving Sub 1-3 (14.4 g, 22.1 mmol) with Toluene (190 mL) in a round bottom flask, Sub 2-31 (5.74 g, 22.1 mmol), Pd.sub.2(dba).sub.3 (0.61 g, 0.66 mmol), P(t-Bu).sub.3 (4.48 g, 22.1 mmol), NaOt-Bu (4.25 g, 44.3 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH.sub.2Cl.sub.2 and water, and the

organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-3 (12.3 g, yield: 73%)

3. Synthesis Example of 1-32

(163) ##STR00165##

(164) After dissolving Sub 1-30 (14.0 g, 21.0 mmol) with Toluene (185 mL) in a round bottom flask, Sub 2-21 (5.99 g, 21.0 mmol), Pd₂(dba)₃ (0.58 g, 0.63 mmol), P(t-Bu)₃ (4.25 g, 21.0 mmol), NaOt-Bu (4.04 g, 42.0 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-32 (12.4 g, yield: 74%)

4. Synthesis Example of 1-35

(165) ##STR00166##

(166) After dissolving Sub 1-33 (12.2 g, 16.1 mmol) with Toluene (160 mL) in a round bottom flask, Sub 2-1 (2.73 g, 16.1 mmol), Pd₂(dba)₃ (0.44 g, 0.48 mmol), P(t-Bu)₃ (3.26 g, 16.1 mmol), NaOt-Bu (3.10 g, 32.2 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-35 (9.80 g, yield: 78%)

5. Synthesis Example of 1-49

(167) ##STR00167##

(168) After dissolving Sub 1-46 (9.77 g, 14.7 mmol) with Toluene (120 mL) in a round bottom flask, Sub 2-1 (2.49 g, 14.7 mmol), Pd₂(dba)₃ (0.40 g, 0.44 mmol), P(t-Bu)₃ (2.97 g, 14.7 mmol), NaOt-Bu (2.83 g, 29.4 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-49 (7.83 g, yield: 78%)

6. Synthesis Example of 1-56

(169) ##STR00168##

(170) After dissolving Sub 1-53 (11.0 g, 16.5 mmol) with Toluene (160 mL) in a round bottom flask, Sub 2-1 (2.80 g, 16.5 mmol), Pd₂(dba)₃ (0.45 g, 0.50 mmol), P(t-Bu)₃ (3.35 g, 16.5 mmol), NaOt-Bu (3.18 g, 33.1 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-56 (8.03 g, yield: 71%)

7. Synthesis Example of 1-89

(171) ##STR00169##

(172) After dissolving Sub 1-81 (15.3 g, 21.3 mmol) with Toluene (200 mL) in a round bottom flask, Sub 2-1 (3.61 g, 21.3 mmol), Pd₂(dba)₃ (0.59 g, 0.64 mmol), P(t-Bu)₃ (4.32 g, 21.3 mmol), NaOt-Bu (4.10 g, 42.7 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-89 (11.7 g, yield: 74%)

8. Synthesis Example of 1-117

(173) ##STR00170##

(174) After dissolving Sub 1-109 (10.9 g, 14.6 mmol) with Toluene (140 mL) in around bottom flask, Sub 2-1 (2.46 g, 14.6 mmol), Pd₂(dba)₃ (0.40 g, 0.44 mmol), P(t-Bu)₃ (2.95 g, 14.6 mmol), NaOt-Bu (2.80 g, 29.1 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH₂Cl₂ and water, and the organic layer was dried over MgSO₄ and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-117 (7.9 g, yield: 76%)

9. Synthesis Example of 1-141

(175) ##STR00171##

(176) After dissolving Sub 1-133 (10.9 g, 14.5 mmol) with Toluene (140 mL) in around bottom flask, Sub 2-33 (4.00 g, 14.5 mmol), Pd.sub.2(dba).sub.3 (0.40 g, 0.43 mmol), P(t-Bu).sub.3 (2.93 g, 14.5 mmol), NaOt-Bu (2.79 g, 29.0 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH.sub.2Cl.sub.2 and water, and the organic layer was dried over MgSO.sub.4 and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-141 (8.69 g, yield: 68%)

10. Synthesis Example of 1-145

(177) ##STR00172##

(178) After dissolving Sub 1-137 (10.3 g, 10.9 mmol) with Toluene (140 mL) in a round bottom flask, Sub 2-1 (1.84 g, 10.9 mmol), Pd.sub.2(dba).sub.3 (0.30 g, 0.33 mmol), P(t-Bu).sub.3 (2.20 g, 10.9 mmol), NaOt-Bu (2.09 g, 21.8 mmol) were added and stirred at 120° C. When the reaction was completed, the resulting compound was extracted with CH.sub.2Cl.sub.2 and water, and the organic layer was dried over MgSO.sub.4 and concentrated, and the resulting compound was recrystallized with a silicagel column to obtain 1-145 (7.50 g, yield: 71%)

(179) Otherwise, FD-MS values of compounds 1-1 to 1-146 of the present invention prepared according to the synthesis example as described above are shown in Table 3 below.

(180) TABLE-US-00004 TABLE 3 compound FD-MS compound FD-MS 1-1 m/z =

685.26(C.sub.48H.sub.35N.sub.3S = 685.89) 1-2 m/z = 775.27(C.sub.54H.sub.37N.sub.3OS =

775.97) 1-3 m/z = 759.29(C.sub.54H.sub.37N.sub.3O.sub.2 = 759.91) 1-4 m/z =

850.31(C.sub.60H.sub.42N.sub.4S = 851.08) 1-5 m/z = 762.28(C.sub.53H.sub.38N.sub.4S =

762.98) 1-6 m/z = 733.26(C.sub.52H.sub.35N.sub.3S = 733.93) 1-7 m/z =

761.29(C.sub.54H.sub.39N.sub.3S = 761.99) 1-8 m/z = 761.29(C.sub.54H.sub.39N.sub.3S =

761.99) 1-9 m/z = 685.26(C.sub.48H.sub.35N.sub.3S = 685.89) 1-10 m/z =

775.27(C.sub.54H.sub.37N.sub.3OS = 775.97) 1-11 m/z = 775.27(C.sub.54H.sub.37N.sub.3OS =

775.97) 1-12 m/z = 801.32(C.sub.57H.sub.43N.sub.3S = 802.05) 1-13 m/z =

719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-14 m/z = 733.26(C.sub.52H.sub.35N.sub.3S =

733.93) 1-15 m/z = 667.26(C.sub.48H.sub.33N.sub.3O = 667.81) 1-16 m/z =

717.28(C.sub.52H.sub.35N.sub.3O = 717.87) 1-17 m/z = 685.26(C.sub.48H.sub.35N.sub.3S =

685.89) 1-18 m/z = 834.34(C.sub.60H.sub.42N.sub.4O = 835.02) 1-19 m/z =

791.24(C.sub.54H.sub.37N.sub.3S.sub.2 = 792.03) 1-20 m/z =

881.25(C.sub.60H.sub.39N.sub.3OS.sub.2 = 882.11) 1-21 m/z =

683.24(C.sub.48H.sub.33N.sub.3S = 683.87) 1-22 m/z = 667.26(C.sub.48H.sub.33N.sub.3O =

667.81) 1-23 m/z = 762.28(C.sub.53H.sub.38N.sub.4S = 762.98) 1-24 m/z =

735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-25 m/z = 821.34(C.sub.60H.sub.43N.sub.3O =

822.02) 1-26 m/z = 710.25(C.sub.49H.sub.34N.sub.4S = 710.90) 1-27 m/z =

686.25(C.sub.47H.sub.34N.sub.4S = 686.88) 1-28 m/z = 669.27(C.sub.48H.sub.35N.sub.3O =

669.83) 1-29 m/z = 901.31(C.sub.64H.sub.43N.sub.3OS = 902.13) 1-30 m/z =

791.24(C.sub.54H.sub.37N.sub.3S.sub.2 = 792.03) 1-31 m/z =

759.29(C.sub.54H.sub.37N.sub.3O.sub.2 = 759.91) 1-32 m/z = 801.32(C.sub.57H.sub.43N.sub.3S

= 802.05) 1-33 m/z = 745.31(C.sub.54H.sub.39N.sub.3O = 745.93) 1-34 m/z =

674.31(C.sub.48H.sub.30D.sub.5N.sub.3O = 674.86) 1-35 m/z =

775.27(C.sub.54H.sub.37N.sub.3OS = 775.97) 1-36 m/z = 923.33(C.sub.67H.sub.45N.sub.3S =

924.18) 1-37 m/z = 834.34(C.sub.60H.sub.42N.sub.4O = 835.02) 1-38 m/z =

735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-39 m/z = 703.25(C.sub.48H.sub.34FN.sub.3S =

703.88) 1-40 m/z = 694.27(C.sub.49H.sub.34N.sub.4O = 694.84) 1-41 m/z =

669.27(C.sub.48H.sub.35N.sub.3O = 669.83) 1-42 m/z = 791.24(C.sub.54H.sub.37N.sub.3S.sub.2

= 792.03) 1-43 m/z = 884.35(C.sub.64H.sub.44N.sub.4O = 885.08) 1-44 m/z =

801.32(C.sub.57H.sub.43N.sub.3S = 802.05) 1-45 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =

735.95) 1-46 m/z = 719.29(C.sub.32H.sub.37N.sub.3O = 719.89) 1-47 m/z =
837.32(C.sub.60H.sub.43N.sub.3S = 838.09) 1-48 m/z = 667.26(C.sub.48H.sub.33N.sub.3O =
667.81) 1-49 m/z = 683.24(C.sub.48H.sub.33N.sub.3S = 683.87) 1-50 m/z =
670.27(C.sub.47H.sub.34N.sub.3O = 670.82) 1-51 m/z = 669.27(C.sub.48H.sub.35N.sub.3O =
669.83) 1-52 m/z = 850.31(C.sub.60H.sub.42N.sub.4S = 851.08) 1-53 m/z =
849.30(C.sub.60H.sub.39N.sub.3O.sub.3 = 850.00) 1-54 m/z =
791.24(C.sub.54H.sub.37N.sub.3S.sub.2 = 792.03) 1-55 m/z = 761.29(C.sub.54H.sub.39N.sub.3S
= 761.99) 1-56 m/z = 683.24(C.sub.48H.sub.33N.sub.3S = 683.87) 1-57 m/z =
667.26(C.sub.48H.sub.33N.sub.3O = 667.81) 1-58 m/z = 683.24(C.sub.48H.sub.33N.sub.3S =
683.87) 1-59 m/z = 667.26(C.sub.48H.sub.33N.sub.3O = 667.81) 1-60 m/z =
715.27(C.sub.49H.sub.37N.sub.3OS = 715.92) 1-61 m/z = 685.26(C.sub.48H.sub.35N.sub.3S =
685.89) 1-62 m/z = 785.34(C.sub.57H.sub.43N.sub.3O = 785.99) 1-63 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-64 m/z = 821.34(C.sub.60H.sub.43N.sub.3O =
822.02) 1-65 m/z = 683.24(C.sub.48H.sub.33N.sub.3S = 683.87) 1-66 m/z =
683.24(C.sub.48H.sub.33N.sub.3S = 683.87) 1-67 m/z = 707.29(C.sub.51H.sub.37N.sub.3O =
707.88) 1-68 m/z = 685.26(C.sub.48H.sub.35N.sub.3S = 685.89) 1-69 m/z =
719.29(C.sub.51H.sub.37N.sub.3O = 719.89) 1-70 m/z = 775.27(C.sub.54H.sub.37N.sub.3OS =
775.97) 1-71 m/z = 731.24(C.sub.49H.sub.37N.sub.3S.sub.2 = 731.98) 1-72 m/z =
735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-73 m/z = 926.34(C.sub.66H.sub.46N.sub.4S =
927.18) 1-74 m/z = 745.31(C.sub.54H.sub.39N.sub.3O = 745.93) 1-75 m/z =
683.24(C.sub.48H.sub.33N.sub.3S = 683.87) 1-76 m/z = 667.26(C.sub.48H.sub.33N.sub.3O =
667.81) 1-77 m/z = 667.26(C.sub.48H.sub.33N.sub.3O = 667.81) 1-78 m/z =
852.33(C.sub.60H.sub.44N.sub.3S = 853.10) 1-79 m/z = 821.34(C.sub.60H.sub.43N.sub.3O =
822.02) 1-80 m/z = 837.32(C.sub.60H.sub.43N.sub.3S = 838.09) 1-81 m/z =
852.33(C.sub.60H.sub.44N.sub.3S = 853.10) 1-82 m/z = 852.33(C.sub.60H.sub.44N.sub.3S =
853.10) 1-83 m/z = 836.35(C.sub.60H.sub.44N.sub.3O = 837.04) 1-84 m/z =
852.33(C.sub.60H.sub.44N.sub.3S = 853.10) 1-85 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-86 m/z = 719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-87 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-88 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-89 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-90 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-91 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-92 m/z = 719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-93 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-94 m/z = 719.29(C.sub.52H.sub.37N.sub.3O =
719.89) 1-95 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-96 m/z =
735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-97 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-98 m/z = 719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-99 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-100 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-101 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-102 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-103 m/z = 719.29(C.sub.52H.sub.37N.sub.3O =
719.89) 1-104 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-105 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-106 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-107 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-108 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-109 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-110 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-111 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-112 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-113 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-114 m/z =
735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-115 m/z = 719.29(C.sub.52H.sub.37N.sub.3O =
719.89) 1-116 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-117 m/z =
719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-118 m/z = 735.27(C.sub.52H.sub.37N.sub.3S =
735.95) 1-119 m/z = 719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-120 m/z =

735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-121 m/z = 719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-122 m/z = 719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-123 m/z = 735.27(C.sub.52H.sub.37N.sub.3S = 735.95) 1-124 m/z = 719.29(C.sub.52H.sub.37N.sub.3O = 719.89) 1-125 m/z = 785.29(C.sub.56H.sub.39N.sub.3S = 786.01) 1-126 m/z = 785.29(C.sub.56H.sub.39N.sub.3S = 786.01) 1-127 m/z = 769.31(C.sub.56H.sub.39N.sub.3O = 769.95) 1-128 m/z = 785.29(C.sub.56H.sub.39N.sub.3S = 786.01) 1-129 m/z = 785.29(C.sub.56H.sub.39N.sub.3S = 786.01) 1-130 m/z = 769.31(C.sub.56H.sub.39N.sub.3O = 769.95) 1-131 m/z = 891.27(C.sub.62H.sub.41N.sub.3S.sub.2 = 892.15) 1-132 m/z = 815.30(C.sub.57H.sub.41N.sub.3OS = 816.04) 1-133 m/z = 837.32(C.sub.60H.sub.43N.sub.3S = 838.09) 1-134 m/z = 745.31(C.sub.54H.sub.39N.sub.3O = 745.93) 1-135 m/z = 761.29(C.sub.54H.sub.39N.sub.3S = 761.99) 1-136 m/z = 1077.38(C.sub.78H.sub.51N.sub.3OS = 1078.35) 1-137 m/z = 887.33(C.sub.64H.sub.45N.sub.3S = 888.15) 1-138 m/z = 911.39(C.sub.67H.sub.49N.sub.3O = 912.15) 1-139 m/z = 1131.45(C.sub.83H.sub.53D.sub.4N.sub.3S = 1132.47) 1-140 m/z = 1018.35(C.sub.70H.sub.46N.sub.6OS = 1019.24) 1-141 m/z = 876.29(C.sub.61H.sub.40N.sub.4OS = 877.08) 1-142 m/z = 1155.41(C.sub.81H.sub.53N.sub.7S = 1156.42) 1-143 m/z = 947.39(C.sub.70H.sub.49N.sub.3O = 948.18) 1-144 m/z = 1049.36(C.sub.74H.sub.49F.sub.2N.sub.3S = 1050.28) 1-145 m/z = 963.36(C.sub.70H.sub.49N.sub.3S = 964.24) 1-146 m/z = 746.30(C.sub.53H.sub.38N.sub.4O = 746.91)

Synthesis Example 2

(181) The compound (final product) represented by Formula 2 according to the present invention may be prepared by reacting Sub 3 and Sub 4 as shown in Scheme 5 below, but is not limited thereto.

(182) ##STR00173##

Synthesis Example of 1'-1

(183) ##STR00174##

(184) After placing Sub 1(1) (34.7 g, 80 mmol) and Sub 2(1) (30.9 g, 80 mmol), K.sub.2CO.sub.3 (19.3 g, 140 mmol), Pd(PPh.sub.3).sub.4 (2.8 g, 2.4 mmol) in a round bottom flask, THF and water were added to dissolve, and then refluxed at 80° C. for 12 hours. When the reaction was completed, the temperature of the reaction product was cooled to room temperature, extracted with CH.sub.2Cl.sub.2, and washed with water. The organic layer was dried over MgSO.sub.4, concentrated, and the resulting organic material was separated using a silicagel column to obtain the desired product (37.4 g, 71%).

Synthesis Example of 1-6

(185) ##STR00175##

(186) Sub 1(6) (44.6 g, 80 mmol) and Sub 2(2) (30.9 g, 80 mmol) were used to obtain a product (43.2 g, 69%) using the synthesis method of 1'-1.

Synthesis Example of 1-12

(187) ##STR00176##

(188) Sub 1(12) (42.7 g, 80 mmol) and Sub 2(33) (34.9 g, 80 mmol) were used to obtain a product (42.7 g, 66%) using the synthesis method of 1'-1.

Synthesis Example of 1-33

(189) ##STR00177##

(190) Sub 1(27) (40.8 g, 80 mmol) and Sub 2(9) (43.1 g, 80 mmol) were used to obtain a product (51.0 g, 72%) using the synthesis method of 1'-1.

Synthesis Example of 1'-44

(191) ##STR00178##

(192) Sub 1(28) (40.8 g, 80 mmol) and Sub 2(10) (37.0 g, 80 mmol) were used to obtain a product (45.4 g, 70%) using the synthesis method of 1'-1.

Synthesis Example of 1-53

(193) ##STR00179##

(194) Sub 1(34) (44.0 g, 80 mmol) and Sub 2(29) (29.6 g, 80 mmol) were used to obtain a product (41.2 g, 68%) using the synthesis method of 1'-1.

Synthesis Example of 1-64

(195) ##STR00180##

(196) Sub 1(37) (48.2 g, 80 mmol) and Sub 2(34) (34.9 g, 80 mmol) were used to obtain a product (45.6 g, 71%) using the synthesis method of 1'-1.

Synthesis Example of 1'-75

(197) ##STR00181##

(198) Sub 1(25) (34.7 g, 80 mmol) and Sub 2(35) (41.8 g, 80 mmol) were used to obtain a product (46.4 g, 73%) using the synthesis method of 1'-1.

Synthesis Example of 2-1

(199) ##STR00182##

(200) Sub 1(1) (34.7 g, 80 mmol) and Sub 2(27) (37.0 g, 80 mmol) were used to obtain a product (42.3 g, 72%) using the synthesis method of 1'-1.

Synthesis Example of 2-22

(201) ##STR00183##

(202) Sub 1(38) (50.7 g, 80 mmol) and Sub 2(24) (37.0 g, 80 mmol) were used to obtain a product (51.6 g, 69%) using the synthesis method of 1'-1.

Synthesis Example of 2-33

(203) ##STR00184##

(204) Sub 1(27) (40.8 g, 80 mmol) and Sub 2(36) (49.2 g, 80 mmol) were used to obtain a product (53.9 g, 70%) using the synthesis method of 1'-1.

Synthesis Example of 2-40

(205) ##STR00185##

(206) Sub 1(25) (34.7 g, 80 mmol) and Sub 2(37) (43.1 g, 80 mmol) were used to obtain a product (44.7 g, 69%) using the synthesis method of 1'-1.

Synthesis Example of 2-51

(207) ##STR00186##

(208) Sub 1(33) (36.0 g, 80 mmol) and Sub 2(38) (35.7 g, 80 mmol) were used to obtain a product (41.1 g, 70%) using the synthesis method of 1'-1.

Synthesis Example of 2-55

(209) ##STR00187##

(210) Sub 1(39) (42.1 g, 80 mmol) and Sub 2(23) (37.0 g, 80 mmol) were used to obtain a product (44.9 g, 68%) using the synthesis method of 1'-1.

Synthesis Example of 2-58

(211) ##STR00188##

(212) Sub 1(25) (34.7 g, 80 mmol) and Sub 2(39) (47.9 g, 80 mmol) were used to obtain a product (45.9 g, 66%) using the synthesis method of 1'-1.

Synthesis Example of 3-10

(213) ##STR00189##

(214) Sub 1(37) (28.6 g, 80 mmol) and Sub 2(10) (37.0 g, 80 mmol) were used to obtain a product (38.9 g, 74%) using the synthesis method of 1'-1.

Synthesis Example of P-41

(215) ##STR00190##

(216) Core 2 (5 g, 14 mmol), Sub 1 (4.6 g, 15.2 mmol), Pd(PPh.sub.3).sub.4 (0.5 g, 0.4 mmol), K.sub.2CO.sub.3 (5.7 g, 41.3 mmol), THF and water were added in a round bottom flask and stirred at 90° C. After the reaction was completed, the reaction mixture was extracted with CH.sub.2Cl.sub.2 and water. The organic layer was dried over MgSO.sub.4 and concentrated. The

resulting compound was separated by silicagel column chromatography and recrystallized to obtain 4.6 g of P-41. (yield: 57%)

Synthesis Example of P-91

(217) ##STR00191##

(218) Core 1 (5 g, 14 mmol), Sub 9 (5.8 g, 15.4 mmol), Pd(PPh.sub.3).sub.4 (0.5 g, 0.4 mmol), K.sub.2CO.sub.3 (5.8 g, 41.9 mmol), THF and water were added in a round bottom flask and stirred at 90° C. After the reaction was completed, the reaction mixture was extracted with CH.sub.2Cl.sub.2 and water. The organic layer was dried over MgSO.sub.4 and concentrated. The resulting compound was separated by silicagel column chromatography and recrystallized to obtain 4.7 g of P-91. (yield: 51%)

Synthesis Example of P-106

(219) ##STR00192##

(220) Core 1 (5 g, 14 mmol), Sub 16 (5.8 g, 15.4 mmol), Pd(PPh.sub.3).sub.4 (0.5 g, 0.4 mmol), K.sub.2CO.sub.3 (5.8 g, 41.9 mmol), THF and water were added in a round bottom flask and stirred at 90° C. After the reaction was completed, the reaction mixture was extracted with CH.sub.2Cl.sub.2 and water. The organic layer was dried over MgSO.sub.4 and concentrated. The resulting compound was separated by silicagel column chromatography and recrystallized to obtain 5.8 g of P-106. (yield: 63%)

Synthesis Example of P-146

(221) ##STR00193##

(222) Core 1 (5 g, 14 mmol), Sub 2 (5.8 g, 15.4 mmol), Pd(PPh.sub.3).sub.4 (0.5 g, 0.4 mmol), K.sub.2CO.sub.3 (5.8 g, 41.9 mmol), THF and water were added in a round bottom flask and stirred at 90° C. After the reaction was completed, the reaction mixture was extracted with CH.sub.2Cl.sub.2 and water. The organic layer was dried over MgSO.sub.4 and concentrated. The resulting compound was separated by silicagel column chromatography and recrystallized to obtain 4.7 g of P-146. (yield: 51%)

Synthesis Example of P-4

(223) ##STR00194##

(224) Core 1 (5 g, 14 mmol), Sub 6 (5.9 g, 15.4 mmol), Pd(PPh.sub.3).sub.4 (0.5 g, 0.4 mmol), K.sub.2CO.sub.3 (5.8 g, 41.9 mmol), THF and water were added in a round bottom flask and stirred at 90° C. After the reaction was completed, the reaction mixture was extracted with CH.sub.2Cl.sub.2 and water. The organic layer was dried over MgSO.sub.4 and concentrated. The resulting compound was separated by silicagel column chromatography and recrystallized to obtain 6.1 g of P-4. (yield: 66%)

Synthesis Example of 4-1

(225) ##STR00195##

(226) Sub 1-1' (50 g, 98.04 mmol) was added to around bottom flask and dissolve with THF (359 mL), Sub 2-1' (52.51 g, 117.65 mmol), Pd(PPh.sub.3).sub.4 (4.53 g, 3.92 mmol), K.sub.2CO.sub.3 (40.65 g, 294.12 mmol) and water (180 mL) were added and stirred to reflux. After the reaction was completed, the reaction mixture was extracted with ether and water. The organic layer was dried over MgSO.sub.4 and concentrated. Thereafter, the concentrate was passed through a silicagel column and recrystallized to obtain 64.61 g of a product. (Yield: 83%)

Synthesis Example of 5-3

(227) ##STR00196##

(228) Sub 1-1" (60 g, 133.35 mmol) was added to a round bottom flask and dissolve with THF (489 mL), Sub 2-3" (58.28 g, 160.01 mmol), Pd(PPh.sub.3).sub.4 (6.16 g, 5.33 mmol), K.sub.2CO.sub.3 (55.29 g, 400.04 mmol), and water (244 mL) were added and stirred to reflux. When the reaction is complete, the resulting compound was extracted with ether and water. The organic layer was dried over MgSO.sub.4 and concentrated. The resulting compound was separated by silicagel column chromatography and recrystallized to obtain 73.40 g of the product. (yield:

75%)

(229) TABLE-US-00005 TABLE 4 compound FD-MS compound FD-MS 1'-1 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 1'-2 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-3 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-4 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 1'-5 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) 1'-6 m/z = 781.22(C.sub.55H.sub.31N.sub.3OS = 781.93) 1'-7 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-8 m/z = 721.22(C.sub.50H.sub.31N.sub.3OS = 721.88) 1'-9 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-10 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-11 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-12 m/z = 807.23(C.sub.57H.sub.33N.sub.3OS = 807.97) 1'-13 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-14 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-15 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-16 m/z = 859.27(C.sub.61H.sub.37N.sub.3OS = 860.05) 1'-17 m/z = 807.23(C.sub.57H.sub.33N.sub.3OS = 807.97) 1'-18 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) 1'-19 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 1'-20 m/z = 859.27(C.sub.61H.sub.37N.sub.3OS = 860.05) 1'-21 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-22 m/z = 857.25(C.sub.61H.sub.35N.sub.3OS = 858.03) 1'-23 m/z = 833.25(C.sub.59H.sub.35N.sub.3OS = 834.01) 1'-24 m/z = 825.23(C.sub.57H.sub.32FN.sub.3OS = 825.96) 1'-25 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 1'-26 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-27 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-28 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-29 m/z = 732.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-30 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-31 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 1'-32 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-33 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 1'-34 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-35 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-36 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-37 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-38 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-39 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-40 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 1'-41 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 1'-42 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-43 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-44 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-45 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-46 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-47 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 1'-48 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 1'-49 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 1'-50 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-51 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-52 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 1'-53 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) 1'-54 m/z = 781.22(C.sub.55H.sub.31N.sub.3OS = 781.93) 1'-55 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 1'-56 m/z = 721.22(C.sub.50H.sub.31N.sub.3OS = 721.88) 1'-57 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-58 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 1'-59 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 1'-60 m/z = 807.23(C.sub.17H.sub.33N.sub.3OS = 807.97) 1'-61 m/z = 723.18(C.sub.49H.sub.29N.sub.3S.sub.2 = 723.91) 1'-62 m/z = 723.18(C.sub.49H.sub.29N.sub.3S.sub.2 = 723.91) 1'-63 m/z = 749.20(C.sub.51H.sub.31N.sub.3S.sub.2 = 749.95) 1'-64 m/z = 875.24(C.sub.61H.sub.37N.sub.3S.sub.2 = 876.11) 1'-65 m/z = 823.21(C.sub.57H.sub.33N.sub.3S.sub.2 = 824.03) 1'-66 m/z = 773.20(C.sub.53H.sub.31N.sub.3S.sub.2 = 773.97) 1'-67 m/z = 799.21(C.sub.55H.sub.33N.sub.3S.sub.2 = 800.01) 1'-68 m/z = 875.24(C.sub.61H.sub.37N.sub.3S.sub.2 = 876.11) 1'-69 m/z =

749.20(C.sub.51H.sub.31N.sub.3S.sub.2 = 749.95) 1'-70 m/z =
873.23(C.sub.61H.sub.35N.sub.3S.sub.2 = 874.09) 1'-71 m/z =
849.23(C.sub.59H.sub.35N.sub.3S.sub.2 = 850.07) 1'-72 m/z =
791.19(C.sub.53H.sub.30FN.sub.3S.sub.2 = 791.96) 1'-73 m/z =
641.21(C.sub.45H.sub.27N.sub.3O.sub.2 = 641.73) 1'-74 m/z =
717.24(C.sub.51H.sub.31N.sub.3O.sub.2 = 717.83) 1'-75 m/z =
798.30(C.sub.57H.sub.30D.sub.5N.sub.3O.sub.2 = 799.0) 1'-76 m/z =
843.29(C.sub.61H.sub.37N.sub.3O.sub.2 = 843.99) 1'-77 m/z =
717.24(C.sub.51H.sub.31N.sub.3O.sub.2 = 717.83) 1'-78 m/z =
717.24(C.sub.51H.sub.31N.sub.3O.sub.3 = 717.83) 1'-79 m/z =
641.21(C.sub.45H.sub.27N.sub.3O.sub.2 = 641.73) 1'-80 m/z =
793.27(C.sub.57H.sub.35N.sub.3O.sub.2 = 793.93) 1'-81 m/z =
869.30(C.sub.63H.sub.39N.sub.3O.sub.2 = 870.02) 1'-82 m/z =
717.24(C.sub.51H.sub.31N.sub.3O.sub.2 = 717.83) 1'-83 m/z =
722.27(C.sub.51H.sub.26D.sub.5N.sub.3O.sub.2 = 722.9) 1'-84 m/z =
793.27(C.sub.57H.sub.35N.sub.3O.sub.2 = 793.93) 2-1 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS
= 733.89) 2-2 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 2-3 m/z =
783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 2-4 m/z = 859.27(C.sub.61H.sub.37N.sub.3OS =
860.05) 2-5 m/z = 833.25(C.sub.59H.sub.35N.sub.3OS = 834.01) 2-6 m/z =
857.25(C.sub.61H.sub.35N.sub.3OS = 858.03) 2-7 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS =
886.09) 2-8 m/z = 797.25(C.sub.56H.sub.35N.sub.3OS = 797.98) 2-9 m/z =
783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 2-10 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS =
783.95) 2-11 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-12 m/z =
883.27(C.sub.63H.sub.37N.sub.3OS = 884.07) 2-13 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS =
783.95) 2-14 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 2-15 m/z =
809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-16 m/z = 935.30(C.sub.67H.sub.41N.sub.3OS =
936.15) 2-17 m/z = 883.27(C.sub.63H.sub.37N.sub.3OS = 884.07) 2-18 m/z =
833.25(C.sub.59H.sub.35N.sub.3OS = 834.01) 2-19 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS =
886.09) 2-20 m/z = 909.28(C.sub.65H.sub.39N.sub.3OS = 910.11) 2-21 m/z =
809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-22 m/z = 933.28(C.sub.67H.sub.39N.sub.3OS =
934.13) 2-23 m/z = 909.28(C.sub.65H.sub.39N.sub.3OS = 910.11) 2-24 m/z =
851.24(C.sub.59H.sub.34FN.sub.3OS = 852.00) 2-25 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS =
733.89) 2-26 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-27 m/z =
885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 2-28 m/z = 935.30(C.sub.67H.sub.41N.sub.3OS =
936.15) 2-29 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-30 m/z =
809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-31 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS =
733.89) 2-32 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 2-33 m/z =
961.31(C.sub.69H.sub.43N.sub.3OS = 962.18) 2-34 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS =
809.99) 2-35 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-36 m/z =
885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 2-37 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS =
886.09) 2-38 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 2-39 m/z =
961.31(C.sub.69H.sub.43N.sub.3OS = 962.18) 2-40 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS =
809.99) 2-41 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 2-42 m/z =
885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 2-43 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS =
886.09) 2-44 m/z = 1032.40(C.sub.74H.sub.54N.sub.3OS = 1033.33) 2-45 m/z =
885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 2-46 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS =
886.09) 2-47 m/z = 961.31(C.sub.69H.sub.43N.sub.3OS = 962.18) 2-48 m/z =
859.27(C.sub.61H.sub.37N.sub.3OS = 860.08) 2-49 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS =
809.99) 2-50 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 2-51 m/z =
733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 2-52 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS =

886.09) 2-53 m/z = 977.29(C.sub.69H.sub.43N.sub.3S.sub.2 = 978.24) 2-54 m/z =
825.23(C.sub.57H.sub.35N.sub.3S.sub.2 = 826.05) 2-55 m/z =
825.23(C.sub.57H.sub.35N.sub.3S.sub.2 = 826.05) 2-56 m/z =
901.26(C.sub.63H.sub.39N.sub.3S.sub.2 = 902.15) 2-57 m/z =
869.30(C.sub.63H.sub.39N.sub.3O.sub.2 = 870.02) 2-58 m/z =
869.30(C.sub.63H.sub.39N.sub.3O.sub.2 = 870.02) 2-59 m/z =
945.34(C.sub.69H.sub.43N.sub.3O.sub.2 = 946.12) 2-60 m/z =
809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 3-1 m/z = 581.16(C.sub.39H.sub.23N.sub.3OS =
581.69) 3-2 m/z = 581.16(C.sub.39H.sub.23N.sub.3OS = 581.69) 3-3 m/z =
581.16(C.sub.39H.sub.23N.sub.3OS = 581.69) 3-4 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS =
783.95) 3-5 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) 3-6 m/z =
731.20(C.sub.51H.sub.29N.sub.3OS = 731.87) 3-7 m/z = 581.16(C.sub.39H.sub.23N.sub.3OS =
581.69) 3-8 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 3-9 m/z =
581.16(C.sub.39H.sub.23N.sub.3OS = 581.69) 3-10 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS =
657.79) 3-11 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 3-12 m/z =
733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 3-13 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS =
631.75) 3-14 m/z = 581.16(C.sub.39H.sub.23N.sub.3OS = 581.69) 3-15 m/z =
581.16(C.sub.39H.sub.23N.sub.3OS = 581.69) 3-16 m/z =
799.21(C.sub.55H.sub.33N.sub.3S.sub.2 = 800.01) 3-17 m/z =
647.15(C.sub.43H.sub.25N.sub.3S.sub.2 = 647.81) 3-18 m/z =
747.18(C.sub.51H.sub.29N.sub.3S.sub.2 = 747.93) 3-19 m/z =
565.18(C.sub.39H.sub.23N.sub.3O.sub.2 = 565.63) 3-20 m/z =
641.21(C.sub.45H.sub.27N.sub.3O.sub.2 = 641.73) 3-21 m/z =
722.27(C.sub.51H.sub.26D.sub.5N.sub.3S.sub.2 = 722.86) 3-22 m/z =
657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) 3-23 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS =
657.79) 3-24 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 3-25 m/z =
733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 3-26 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS =
707.85) 3-27 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) 3-28 m/z =
859.27(C.sub.61H.sub.37N.sub.3OS = 860.05) 3-29 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS =
657.79) 3-30 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) 3-31 m/z =
809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 3-32 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS =
809.99) 3-33 m/z = 885.28(C.sub.63H.sub.39N.sub.3OS = 886.09) 3-34 m/z =
809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) 3-35 m/z =
673.16(C.sub.45H.sub.27N.sub.3S.sub.2 = 673.85) 3-36 m/z =
793.27(C.sub.57H.sub.35N.sub.3O.sub.2 = 793.93) P-1 m/z = 581.16(C.sub.39H.sub.23N.sub.3OS
= 581.69) P-2 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) P-3 m/z =
631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) P-4 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS =
657.79) P-5 m/z = 681.19(C.sub.47H.sub.27N.sub.3OS = 681.81) P-6 m/z =
631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) P-7 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS=
631.75) P-8 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) P-9 m/z =
657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) P-10 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS =
657.79) P-11 m/z = 586.19(C.sub.39H.sub.18D.sub.5N.sub.3OS = 586.7) P-12 m/z =
681.19(C.sub.47H.sub.27N.sub.3OS = 681.81) P-13 m/z = 681.19(C.sub.47H.sub.27N.sub.3OS =
681.81) P-14 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-15 m/z =
707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-16 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS =
707.85) P-17 m/z = 636.20(C.sub.43H.sub.20D.sub.5N.sub.3OS = 636.8) P-18 m/z =
681.19(C.sub.47H.sub.27N.sub.3OS = 681.81) P-19 m/z = 681.19(C.sub.47H.sub.27N.sub.3OS =
681.81) P-20 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-21 m/z =
707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-22 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS =
707.85) P-23 m/z = 636.20(C.sub.43H.sub.20D.sub.5N.sub.3OS = 636.8) P-24 m/z =

707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-25 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-26 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-27 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-28 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-29 m/z = 662.22(C.sub.45H.sub.22D.sub.5N.sub.3OS = 662.8) P-30 m/z = 662.22(C.sub.45H.sub.22D.sub.5N.sub.3OS = 662.82) P-31 m/z = 581.16(C.sub.39H.sub.23N.sub.3OS = 581.69) P-32 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) P-33 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) P-34 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) P-35 m/z = 681.19(C.sub.47H.sub.27N.sub.3OS = 681.81) P-36 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) P-37 m/z = 631.17(C.sub.43H.sub.25N.sub.3OS = 631.75) P-38 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) P-39 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) P-40 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) P-41 m/z = 586.19(C.sub.39H.sub.18D.sub.5N.sub.3OS = 586.7) P-42 m/z = 681.19(C.sub.47H.sub.27N.sub.3OS = 681.81) P-43 m/z = 681.19(C.sub.47H.sub.27N.sub.3OS = 681.81) P-44 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-45 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-46 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-47 m/z = 636.20(C.sub.43H.sub.20D.sub.5N.sub.3OS = 636.8) P-48 m/z = 681.19(C.sub.47N.sub.27N.sub.3OS = 681.81) P-49 m/z = 681.19(C.sub.47H.sub.27N.sub.3OS = 681.81) P-50 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-51 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-52 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-53 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-54 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-55 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-56 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-57 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-58 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-59 m/z = 662.22(C.sub.45H.sub.22DN.sub.3OS = 662.82) P-60 m/z = 662.22(C.sub.45H.sub.22DN.sub.3OS = 662.82) P-61 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS = 657.79) P-62 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-63 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-64 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-65 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) P-66 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-67 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-68 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-69 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-70 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-71 m/z = 662.22(C.sub.45H.sub.22D.sub.5N.sub.3OS = 662.8) P-72 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) P-73 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) P-74 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-75 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-76 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-77 m/z = 712.23(C.sub.49H.sub.24D.sub.5N.sub.3OS = 712.9) P-78 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) P-79 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) P-80 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-81 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-82 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-83 m/z = 712.23(C.sub.49H.sub.24D.sub.5N.sub.3OS = 712.9) P-84 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-85 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-86 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) P-87 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) P-88 m/z = 809.25(C.sub.57H.sub.35N.sub.3OS = 809.99) P-89 m/z = 738.25(C.sub.51H.sub.26D.sub.5N.sub.3OS = 738.9) P-90 m/z = 738.25(C.sub.51H.sub.26D.sub.5N.sub.3OS = 738.92) P-91 m/z = 662.22(C.sub.45H.sub.22DN.sub.3OS = 662.82) P-92 m/z = 712.23(C.sub.49H.sub.24D.sub.5N.sub.3OS = 712.88) P-93 m/z = 712.23(C.sub.49H.sub.24D.sub.5N.sub.3OS = 712.9) P-94 m/z = 738.25(C.sub.51H.sub.26D.sub.5N.sub.3OS = 738.92) P-95 m/z =

762.25(C.sub.53H.sub.26D.sub.5N.sub.3OS = 762.9) P-96 m/z =
712.23(C.sub.49H.sub.24D.sub.5N.sub.3OS = 712.88) P-97 m/z =
712.23(C.sub.49H.sub.24D.sub.5N.sub.3OS = 712.9) P-98 m/z =
738.25(C.sub.51H.sub.26D.sub.5N.sub.3OS = 738.92) P-99 m/z =
738.25(C.sub.51H.sub.26D.sub.5N.sub.3OS = 738.9) P-100 m/z =
738.25(C.sub.51H.sub.26D.sub.5N.sub.3OS = 738.92) P-101 m/z =
667.2(C.sub.45H.sub.17D.sub.10N.sub.3OS = 667.9) P-102 m/z =
707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-103 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS =
707.85) P-104 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-105 m/z =
733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-106 m/z = 657.19(C.sub.45H.sub.27N.sub.3OS =
657.79) P-107 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-108 m/z =
707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-109 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS =
733.89) P-110 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) P-111 m/z =
707.20(C.sub.49H.sub.29N.sub.3OS = 707.85) P-112 m/z = 707.20(C.sub.49H.sub.29N.sub.3OS =
707.85) P-113 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-114 m/z =
733.22(C.sub.51H.sub.31N.sub.3OS = 733.89) P-115 m/z = 733.22(C.sub.51H.sub.31N.sub.3OS =
733.89) P-116 m/z = 662.22(C.sub.45H.sub.22D.sub.5N.sub.3OS = 662.82) P-117 m/z =
757.22(C.sub.53H.sub.31N.sub.3OS = 757.91) P-118 m/z = 757.22(C.sub.53H.sub.31N.sub.3OS =
757.91) P-119 m/z = 783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) P-120 m/z =
783.23(C.sub.55H.sub.33N.sub.3OS = 783.95) 4-1 m/z = 793.27 (C.sub.57H.sub.35N.sub.3O.sub.2
= 793.93) 4-2 m/z = 869.30 (C.sub.63H.sub.39N.sub.3O.sub.2 = 870.02) 4-4 m/z =
798.3(C.sub.57H.sub.30D.sub.5N.sub.3O2 = 798.96) 4-7 m/z = 919.32
(C.sub.67H.sub.41N.sub.3O.sub.2 = 920.08) 4-8 m/z = 843.29 (C.sub.61H.sub.37N.sub.3O.sub.2 =
843.99) 4-10 m/z = 945.34 (C.sub.69H.sub.43N.sub.3O.sub.2 = 946.12) 4-11 m/z = 944.32
(C.sub.68H.sub.40N.sub.4O.sub.2 = 945.09) 4-12 m/z = 970.33 (C.sub.70H.sub.42N.sub.4O.sub.2
= 971.13) 5-1 m/z = 657.19 (C.sub.45H.sub.27N.sub.3OS = 657.79) 5-2 m/z = 733.22
(C.sub.51H.sub.31N.sub.3OS = 733.89) 5-3 m/z = 733.22 (C.sub.51H.sub.31N.sub.3OS = 733.89)
5-4 m/z = 733.22 (C.sub.51H.sub.31N.sub.3OS = 733.89) 5-7 m/z = 707.20
(C.sub.49H.sub.29N.sub.3OS = 707.85) 5-8 m/z = 757.22 (C.sub.53H.sub.31N.sub.3OS = 757.91)
5-9 m/z = 859.27 (C.sub.61H.sub.37N.sub.3OS = 860.05) 5-10 m/z = 707.20
(C.sub.49H.sub.29N.sub.3OS = 707.85) 5-11 m/z = 808.23 (C.sub.56H.sub.32N.sub.4OS = 808.96)
5-12 m/z = 890.31 (C.sub.63H.sub.34D.sub.5N.sub.3OS = 891.1) 5-13 m/z = 824.24
(C.sub.55H.sub.32N.sub.6OS) = 824.96) 5-14 m/z = 752.20 (C.sub.50H.sub.29FN.sub.4OS =
752.87) 5-15 m/z = 765.26 (C.sub.51H.sub.35N.sub.5OS = 765.94) 5-16 m/z = 765.17
(C.sub.49H.sub.27N.sub.5OS.sub.2 = 765.91) 5-17 m/z = 807.23 (C.sub.57H.sub.33N.sub.3OS =
807.97) 5-18 m/z = 833.25 (C.sub.59H.sub.35N.sub.3OS = 834.01) 5-19 m/z = 733.22
(C.sub.51H.sub.31N.sub.3OS = 733.89) 5-20 m/z = 733.22 (C.sub.51H.sub.31N.sub.3OS =
733.89)

(230) Otherwise, the synthesis examples of the present invention represented by the Formulas 1 and 2 have been described, but these are all based on the Buchwald-Hartwig cross coupling reaction, Miyaura boration reaction, Suzuki cross-coupling reaction, Intramolecular acid-induced cyclization reaction (*J. mater. Chem.* 1999, 9, 2095.), Pd(II)-catalyzed oxidative cyclization reaction (*Org. Lett.* 2011, 13, 5504), and PPh.sub.3-mediated reductive cyclization reaction (*J. Org. Chem.* 2005, 70, 5014.), and those skilled in the art will readily understand that the above reaction proceeds even when, besides the substituent specified in the specific synthesis example, other substituents (X.sup.1 to X.sup.3, L.sup.1 to L.sup.7, Ar.sup.1 to Ar.sup.8, Substituents such as A, B, C, D, E, F, G and H) defined in the Formulas 1 and 2 are bonded.

(231) Evaluation of Manufacture of Organic Electric Element

Example 1) Manufacture and Evaluation of Green Organic Light Emitting Diode

(232) First, on an ITO layer(anode) formed on a glass substrate, N.sup.1-(naphthalen-2-yl)-

4.1 12.8 5000.0 39.2 131.5 0.33 0.63 example30 1-28 4.0 13.1 5000.0 38.1 130.8 0.34 0.65
example31 1-30 4.0 12.6 5000.0 39.6 124.4 0.33 0.63 example32 1-35 3.9 12.9 5000.0 38.8 132.2
0.31 0.64 example33 1-38 4.0 13.3 5000.0 37.6 128.7 0.30 0.65 example34 1-39 4.1 12.2 5000.0
40.9 128.7 0.32 0.61 example35 1-40 4.2 12.7 5000.0 39.4 122.2 0.34 0.62 example36 1-42 4.1
12.1 5000.0 41.2 123.6 0.33 0.63 example37 1-48 4.2 12.2 5000.0 41.1 127.8 0.31 0.65 example38
1-102 4.3 12.3 5000.0 40.5 126.9 0.31 0.64 example39 1-143 4.2 11.8 5000.0 42.4 126.9 0.31 0.64
example40 1-1 P-69 4.1 12.5 5000.0 39.9 130.1 0.33 0.60 example41 1-4 4.1 12.8 5000.0 39.1
130.5 0.31 0.60 example42 1-6 4.0 12.4 5000.0 40.5 130.2 0.33 0.61 example43 1-28 4.0 13.1
5000.0 38.0 133.3 0.34 0.62 example44 1-30 4.1 12.6 5000.0 39.6 124.1 0.35 0.62 example45 1-35
4.1 12.8 5000.0 39.0 133.3 0.30 0.62 example46 1-38 4.0 12.8 5000.0 39.1 131.6 0.35 0.64
example47 1-39 4.2 12.4 5000.0 40.4 129.5 0.32 0.63 example48 1-40 4.2 12.4 5000.0 40.5 120.4
0.31 0.62 example49 1-42 4.1 12.3 5000.0 40.8 121.8 0.32 0.61 example50 1-48 4.1 12.0 5000.0
41.7 120.9 0.35 0.61 example51 1-102 4.2 12.7 5000.0 39.5 128.7 0.34 0.61 example52 1-143 4.2
11.8 5000.0 42.2 126.4 0.33 0.63 example53 1-1 4-5 4.0 12.4 5000.0 40.2 126.2 0.31 0.64
example54 1-4 4.0 12.5 5000.0 40.1 131.2 0.33 0.61 example55 1-6 4.0 13.0 5000.0 38.6 133.4
0.31 0.62 example56 1-28 4.0 12.7 5000.0 39.3 134.2 0.34 0.62 example57 1-30 3.9 12.7 5000.0
39.5 127.3 0.35 0.61 example58 1-35 4.0 13.3 5000.0 37.6 131.7 0.30 0.64 example59 1-38 3.9
12.8 5000.0 39.1 132.9 0.35 0.64 example60 1-39 4.2 12.1 5000.0 41.3 125.8 0.31 0.65 example61
1-40 4.2 12.8 5000.0 38.9 126.8 0.32 0.60 example62 1-42 4.1 12.1 5000.0 41.4 120.4 0.33 0.63
example63 1-48 4.2 11.9 5000.0 41.8 121.4 0.34 0.64 example64 1-102 4.2 12.6 5000.0 39.6 124.6
0.32 0.61 example65 1-143 4.2 11.9 5000.0 42.0 125.6 0.30 0.61 example53 1-1 5-2 4.1 12.6
5000.0 39.7 127.9 0.32 0.62 example54 1-4 4.1 12.6 5000.0 39.7 128.9 0.34 0.60 example55 1-6
4.1 12.6 5000.0 39.8 131.4 0.34 0.61 example56 1-28 4.1 12.7 5000.0 39.4 133.1 0.33 0.65
example57 1-30 4.0 12.4 5000.0 40.4 125.0 0.34 0.64 example58 1-35 4.0 13.0 5000.0 38.4 130.6
0.34 0.65 example59 1-38 3.9 12.8 5000.0 39.1 131.4 0.35 0.63 example60 1-39 4.2 12.5 5000.0
39.9 127.5 0.30 0.63 example61 1-40 4.1 12.7 5000.0 39.4 124.1 0.31 0.60 example62 1-42 4.1
12.1 5000.0 41.5 125.7 0.34 0.61 example63 1-48 4.2 12.2 5000.0 40.8 127.4 0.34 0.61 example64
1-102 4.2 12.3 5000.0 40.5 125.5 0.32 0.63 example65 1-143 4.3 11.8 5000.0 42.2 123.9 0.34 0.63

(238) As can be seen from the results of Table 5, when the material for an organic electroluminescent device of the present invention represented by Formula 1 and Formula 2 is mixed and used as a phosphorescent host (Examples 1 to 56), compared to devices using a single compound (Comparative Examples 1 to 3) or device mixed with a comparative compound (Comparative Examples 4 to 6), the driving voltage, efficiency, and lifespan are significantly improved.

(239) First, it can be seen that driving, efficiency, and lifespan increase as the number of amines substituted in the core increases among the comparative compounds having the same dibenzofuran core. That is, compared to Comparative Example 1 in which three carbazoles were substituted in dibenzofuran, the device results in Comparative Examples 2 and 3 in which an amine group was substituted in the same core showed excellent electrical properties, and Compared to Comparative Example 2 in which one amine group was substituted on the same core, Comparative Example 3 device in which two amine groups were substituted showed superior electrical characteristics. At this time, in the case of Comparative Examples 4 to 6 using a mixture of Comparative Compounds A to C and the compound represented by Formula 2 as a phosphorescent host, all electrical characteristics of the devices of Comparative Examples 1 to 3 using a single material were improved. It can be seen that even a compound having poor device performance with a single host can improve the electrical characteristics of the device when mixed with a compound having a good charge balance.

(240) It can be seen that Examples 1 to 65 in which the compounds of Formula 1 and Formula 2 of the present invention were mixed and used as a host were significantly improved than in Comparative Examples 3 to 6.

(241) Based on the above experimental results, the present inventors determined that in the case of a mixture of the compound of Formula 1 and the compound of Formula 2, each of the compounds has novel properties other than those of the compound, and measured the PL lifetime using the compound of Formula 1, the compound of Formula 2, and the mixture of the present invention, respectively. As a result, it was confirmed that when the compounds of the present invention, Formula 1 and Formula 2, were mixed, a new PL wavelength was formed unlike the single compound, and the decrease and disappearance time of the newly formed PL wavelength increased from about 60 times to about 360 times less than the decrease and disappearance time of each of the compounds of Formula 1 and Formula 2. It is considered when mixed with the compound of the present invention, not only electrons and holes are moved through the energy level of each compound, but also the efficiency and life span are increased by electron, hole transport or energy transfer by a new region(exciplex) having a new energy level formed due to mixing. As a result, when the mixture of the present invention is used, the mixed thin film is an important example showing exciplex energy transfer and light emitting process.

(242) Also, the reason why the combination of the present invention is superior to Comparative Examples 4 to 6 in which a comparative compound is mixed and used as a phosphorescent host is that Formula 1 with high hole transport and stability by introducing Dibenzothiophene or Dibenzofuran between amine groups has a good electrochemical synergy effect with Compound represented by Formula 2 with strong electron properties. Accordingly, the charge balance between holes and electrons in the emission layer is increased, so that light emission is well performed inside the emitting layer rather than the hole transport layer interface, thereby reducing deterioration at the HTL interface, maximizing the driving voltage, efficiency, and lifespan of the entire device. That is, in conclusion, it is considered that the combination of Formula 1 and Formula 2 has an electrochemical synergy effect to improve the performance of the entire device.

Example 2) Manufacture and Evaluation of Green Organic Light Emitting Diode by Mixing Ratio

(243) TABLE-US-00007 TABLE 6 Mixing ratio (first Current First Second host:second Density Brightness Efficiency host host host) Voltage (mA/cm.sup.2) (cd/m.sup.2) (cd/A) T(95) example66 1-1 1'-25 7:3 4.1 12.2 5000.0 41.1 133.5 example67 5:5 3.9 12.6 5000.0 39.7 131.1 example68 4:6 3.8 12.9 5000.0 38.6 127.7 example69 3:7 4.1 13.6 5000.0 36.8 124.5 example70 1-40 P-26 7:3 4.1 12.2 5000.0 40.8 132.5 example71 5:5 3.9 12.7 5000.0 39.2 129.3 example72 4:6 3.8 12.8 5000.0 38.9 128.6 example73 3:7 4.0 13.7 5000.0 36.5 125.3

(244) As shown in Table 6, a device was manufactured and measured in the same manner as in Example 1 by using a mixture of the compounds of the present invention in different ratios (7:3, 5:5, 4:6, 3:7). As a result of measuring by ratio, in the case of 7:3, it was similar to the result of Example 1, which was measured as 6:4, but in the case of 5:5, 4:6, and 3:7 where the ratio of the first host decreases, the results of driving voltage, efficiency, and lifespan gradually declined. This can be explained because when an appropriate amount of the compound represented by Formula 1 having strong hole properties such as 7:3 and 4:6 is mixed, the charge balance in the emitting layer is maximized.

(245) Although exemplary embodiments of the present invention have been described for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the invention as disclosed in the accompanying claims. Therefore, the embodiment disclosed in the present invention is intended to illustrate the scope of the technical idea of the present invention, and the scope of the present invention is not limited by the embodiment.

(246) The scope of the present invention shall be construed on the basis of the accompanying claims, and it shall be construed that all of the technical ideas included within the scope equivalent to the claims belong to the present invention.

INDUSTRIAL AVAILABILITY

(247) According to the present invention, it is possible to manufacture an organic device having

excellent device characteristics of high luminance, high light emission and long life, and thus has industrial applicability.

Claims

1. An organic electronic element comprising a first electrode, a second electrode, and an organic material layer formed between the first electrode and the second electrode, wherein the organic material layer comprises an emitting layer, wherein the emitting layer comprises a first host compound represented by Formula 1 and a second host compound represented by Formula 2 as a phosphorescent emitting layer: ##STR00198## wherein: 1) A and B rings are each independently a C.sub.6-C.sub.20 aryl or C.sub.2-C.sub.20 heterocycle group, 2) X.sup.1 is S or O, 3) X.sup.2 is N-L.sup.7-Ar.sup.9, O, S or CR'R'', wherein R' and R'' are each independently hydrogen; a C.sub.6-C.sub.60 aryl group; a fluorenyl group; or a C.sub.1-C.sub.50 alkyl group, and R' and R'' may be bonded to each other to form a spiro ring, 4) p and q are each an integer of 0-10, r is an integer of 0-3, s is an integer of 0-4, 5) R.sup.1, R.sup.2, R.sup.3 and R.sup.4 are each independently selected from the group consisting of hydrogen; a C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60 heterocyclic group including at least one hetero atom of O, N, S, Si or P; a C.sub.1-C.sub.50 alkyl group; a C.sub.2-C.sub.20 alkenyl group; a C.sub.2-C.sub.20 alkynyl group; and a C.sub.1-C.sub.30 alkoxy group; 6) Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, Ar.sup.7, Ar.sup.8 and Ar.sup.9 are each independently selected from the group consisting of a C.sub.6-C.sub.60 aryl group; a C.sub.2-C.sub.60 heterocyclic group including at least one hetero atom of O, N, S, Si or P; a fluorenyl group; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.1-C.sub.50 alkyl group; and a C.sub.2-C.sub.20 alkenyl group; 7) L.sup.1, L.sup.2, L.sup.3, L.sup.4, L.sup.5, L.sup.5, L.sup.6 and L.sup.7 are independently selected from the group consisting of a single bond; a C.sub.6-C.sub.60 arylene group; a fluorenylene group; and a C.sub.2-C.sub.60 heteroarylene group containing at least one hetero atom of O, N, S, Si or P, wherein, the aryl group, fluorenyl group, arylene group, heterocyclic group, fluorenylene group, fused ring group, alkyl group, alkenyl group, alkoxy group and aryloxy group may be substituted with one or more substituents selected from the group consisting of deuterium; halogen; a cyano group; nitro group; C.sub.1-C.sub.20 alkoxy group; C.sub.1-C.sub.20 alkyl group; C.sub.2-C.sub.20 alkenyl group; C.sub.2-C.sub.20 alkynyl group; C.sub.6-C.sub.20 aryl group; C.sub.6-C.sub.20 aryl group substituted with deuterium; a fluorenyl group; C.sub.2-C.sub.20 heterocyclic group; and C.sub.3-C.sub.20 cycloalkyl group; wherein the substituents may be bonded to each other to form a saturated or unsaturated ring, wherein the term 'ring' means a C.sub.3-C.sub.60 aliphatic ring or a C.sub.6-C.sub.60 aromatic ring or a C.sub.2-C.sub.60 heterocyclic group or a fused ring formed by the combination thereof.

2. The organic electronic element of claim 1, wherein A or B in Formula 1 is each independently selected from the group consisting of Formulas a-1 to a-7: ##STR00199## wherein: Z.sup.1 to Z.sup.48 are each independently CR.sup.c or N, Z.sup.1 to Z.sup.48 bonded to L.sup.1 to L.sup.7 are carbon (C), R.sup.c is selected from the group consisting of a C.sub.5-C.sub.50 aryl group; a fluorenyl group; a fused ring group of a C.sub.3-C.sub.60 alicyclic ring and a C.sub.5-C.sub.50 aromatic ring, a C.sub.2-C.sub.20 heterocyclic group including at least one hetero atom of O, N, S, Si or P, * indicates the position to be condensed.

3. The organic electronic element of claim 1, wherein L.sup.1, L.sup.2, L.sup.3, L.sup.4, L.sup.5, L.sup.6 and L.sup.7 in Formula 1 or Formula 2 are represented by one of Formulas b-1 to b-13: ##STR00200## ##STR00201## ##STR00202## wherein: Y is N-L.sup.8-Ar.sup.10, O, S or CR'R'', L.sup.8 is the same as the definition of L.sup.1 in claim 1, Ar.sup.10 is as the same as the definition of Ar.sup.1 in claim 1, R' and R'' are the same as defined in claim 1, a, c, d and e are each independently an integer of 0 to 4, and b is an integer of 0 to 6, f and g are each independently an integer of 0 to 3, h is an integer of 0 to 2, i is an integer of 0 or 1, R.sup.5, R.sup.6 and R.sup.7 are

each independently hydrogen; deuterium; tritium; halogen; cyano group; nitro group; C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60 heterocyclic group including at least one heteroatom of O, N, S, Si or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.1-C.sub.50 alkyl group; a C.sub.2-C.sub.20 alkenyl group; a C.sub.2-C.sub.20 alkynyl group; a C.sub.1-C.sub.30 alkoxy group; a C.sub.6-C.sub.30 aryloxy group; and -L.sup.a-N(R.sup.d)(R.sup.e); or in case a, b, c, d, e, f and g are 2 or more, and h is 2 or more, R.sup.5, R.sup.6 and R.sup.7 are in plural being the same as or different, and a plurality of R.sup.5 or a plurality of R.sup.6 or a plurality of R.sup.7 or adjacent R.sup.5 and R.sup.6, or adjacent R.sup.6 and R.sup.7 may be bonded to each other to form an aromatic or a heteroaromatic ring, wherein L.sup.a is selected from the group consisting of a single bond; a C.sub.6-C.sub.60 arylene group; a fluorenylene group; a C.sub.2-C.sub.60 heterocyclic group containing at least one hetero atom of O, N, S, Si, or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.3-C.sub.60 aromatic ring; and C.sub.3-C.sub.60 aliphatic hydrocarbon group; R.sup.d and R.sup.e are each independently selected from the group consisting of a C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60 heterocyclic group containing at least one hetero atom of O, N, S, Si, or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring, Z.sup.49, Z.sup.50, and Z.sup.51 are each independently CR.sup.g or N, at least one of Z.sup.49, Z.sup.50, and Z.sup.51 is N, R.sup.g is selected from the group consisting of hydrogen; deuterium; tritium; halogen; cyano group; nitro group; C.sub.6-C.sub.60 aryl group; a fluorenyl group; a C.sub.2-C.sub.60 heterocyclic group including at least one heteroatom of O, N, S, Si or P; a fused ring group of a C.sub.3-C.sub.60 aliphatic ring and a C.sub.6-C.sub.60 aromatic ring; a C.sub.1-C.sub.50 alkyl group; a C.sub.2-C.sub.20 alkenyl group; a C.sub.2-C.sub.20 alkynyl group; a C.sub.1-C.sub.30 alkoxy group; a C.sub.6-C.sub.30 aryloxy group; and adjacent R.sup.5 and R.sup.g may be bonded to each other to form an aromatic or a heteroaromatic ring.

4. The organic electronic element of claim 1, wherein at least one of Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4 and Ar.sup.5 is represented by Formula 1-2: ##STR00203## wherein: C and D are the same as the definition of A in claim 1, X.sup.3 is N-L.sup.10-Ar.sup.11, O, S or CR'R", L.sup.9 and L.sup.10 are the same as the definition of L.sup.1 in claim 1, Ar.sup.11 is the same as the definition of Ar.sup.1 in claim 1, R' and R" are the same as defined in claim 1.

5. The organic electronic element of claim 1, wherein the first host compound represented by Formula 1 includes a compound represented by Formula 3 or Formula 4: ##STR00204## wherein: Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, L.sup.1, L.sup.2, L.sup.3, R.sup.1, R.sup.2 are the same as defined in claim 1, p' is an integer of 0 to 3, and q' is an integer of 0 to 2.

6. The organic electronic element of claim 1, wherein the first host compound represented by Formula 1 includes a compound represented by any of Formulas 5 to 11: ##STR00205## ##STR00206## wherein: L.sup.1, L.sup.2, L.sup.3, Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, R.sup.1, R.sup.2, X.sup.1 are the same as defined in claim 1, p' is an integer of 0 to 3, and q' is an integer of 0 to 2, o is an integer of 0 to 4.

7. The organic electronic element of claim 1, wherein the first host compound represented by Formula 1 includes a compound represented by any of Formulas 12 to 21: ##STR00207## ##STR00208## wherein: Ar.sup.1, Ar.sup.2, Ar.sup.3, Ar.sup.4, Ar.sup.5, Ar.sup.6, L.sup.1, L.sup.2, L.sup.3, R.sup.1, R.sup.2, p, q, X.sup.1, A and B are the same as defined in claim 1, p' is an integer of 0 to 3, and q' is an integer of 0 to 2.

8. The organic electronic element of claim 1, wherein the compound represented by Formula 1 includes any of the following compounds: ##STR00209## ##STR00210## ##STR00211## ##STR00212## ##STR00213## ##STR00214## ##STR00215## ##STR00216## ##STR00217## ##STR00218## ##STR00219## ##STR00220## ##STR00221## ##STR00222## ##STR00223## ##STR00224## ##STR00225## ##STR00226## ##STR00227## ##STR00228## ##STR00229## ##STR00230## ##STR00231##

9. The organic electronic element of claim 1, wherein the second host compound represented by

Formula 2 is represented by any of Formulas 22 to 25: ##STR00232## wherein: X.sup.2, L.sup.4, L.sup.5, L.sup.6, Ar.sup.7, Ar.sup.8, R.sup.3, R.sup.4, r and s are the same as defined in claim 1.

10. The organic electronic element of claim 1, wherein the second host compound represented by Formula 2 includes a compound represented by Formula 26: ##STR00233## wherein: X.sup.2, L.sup.4, L.sup.5, L.sup.6, Ar.sup.7, R.sup.3, R.sup.4, r and s are the same as defined in claim 1, X.sup.4 is the same as the definition of X.sup.2 in claim 1, R.sup.8 and R.sup.9 are the same as the definition of R.sup.3 and R.sup.4 in claim 1, u is the same as the definition of r in claim 1, and t is the same as the definition of s in claim 1.

11. The organic electronic element of claim 1, wherein the second host compound represented by Formula 2 includes a compound represented by any of Formulas 27 to 30: ##STR00234## wherein: X.sup.2, L.sup.4, L.sup.5, L.sup.6, Ar.sup.7, R.sup.3, R.sup.4, r and s are the same as defined in claim 1, X.sup.4 is the same as the definition of X.sup.2 in claim 1, R.sup.8 and R.sup.9 are the same as definition of R.sup.3 and R.sup.4 in claim 1, u is the same as the definition of r in claim 1, and t is the same as the definition of s in claim 1.

12. The organic electronic element of claim 1, wherein the second host compound represented by Formula 2 includes any of the following compounds: ##STR00235## ##STR00236## ##STR00237## ##STR00238## ##STR00239## ##STR00240## ##STR00241## ##STR00242## ##STR00243## ##STR00244## ##STR00245## ##STR00246## ##STR00247## ##STR00248## ##STR00249## ##STR00250## ##STR00251## ##STR00252## ##STR00253## ##STR00254## ##STR00255## ##STR00256## ##STR00257## ##STR00258## ##STR00259## ##STR00260## ##STR00261## ##STR00262## ##STR00263## ##STR00264## ##STR00265## ##STR00266## ##STR00267## ##STR00268## ##STR00269## ##STR00270## ##STR00271## ##STR00272## ##STR00273## ##STR00274## ##STR00275## ##STR00276## ##STR00277## ##STR00278## ##STR00279## ##STR00280## ##STR00281## ##STR00282## ##STR00283## ##STR00284## ##STR00285## ##STR00286## ##STR00287## ##STR00288## ##STR00289## ##STR00290## ##STR00291## ##STR00292## ##STR00293## ##STR00294## ##STR00295## ##STR00296## ##STR00297## ##STR00298## ##STR00299## ##STR00300## ##STR00301## ##STR00302## ##STR00303## ##STR00304## ##STR00305## ##STR00306## ##STR00307## ##STR00308## ##STR00309## ##STR00310## ##STR00311## ##STR00312## ##STR00313## ##STR00314## ##STR00315## ##STR00316## ##STR00317## ##STR00318## ##STR00319## ##STR00320## ##STR00321## ##STR00322## ##STR00323##

13. The organic electronic element of claim 1 comprising a hole transport layer, an emitting auxiliary layer, or both the layers between the first electrode and the emitting layer, wherein either of or both the layers include the compound of Formula 1.

14. The organic electronic element of claim 1, wherein the compounds represented by Formula 1 and by Formula 2 are mixed in a ratio of 1:9 to 9:1 and are included in the emitting layer.

15. The organic electronic element of claim 1, wherein the compounds represented by Formula 1 and by Formula 2 are mixed in a ratio of 1:9 to 5:5 and are included in the emitting layer.

16. The organic electronic element of claim 1, wherein the compounds represented by Formula 1 and by Formula 2 are mixed in a ratio of 2:8 or 3:7 and are included in the emitting layer.

17. An electronic device comprising: a display device including the organic electric element of claim 1; and a control unit for driving the display device.

18. The compound of claim 17, wherein the organic electric element is at least one of an OLED, an organic solar cell, an organic photo conductor, an organic transistor and an element for monochromic or white illumination.
